

State dependent government spending multipliers: Downward nominal wage rigidity and sources of business cycle fluctuations*

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Abstract

In a New Keynesian model with downward nominal wage rigidity (DNWR), we show that government spending is more effective in stimulating output in a low-inflation recession relative to a high-inflation recession. The government spending multiplier is large when DNWR binds, but the nature of recession matters due to the opposing response of inflation, and consequently for real wages. Using U.S. historical time series data, we provide evidence of larger spending multipliers in low inflation recessions and the importance of the depth of recessions. We also employ cross-sectional data from U.S. states to document supporting evidence on multipliers and our proposed mechanism.

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1 Introduction

The recent periods of low interest rates have shown that fiscal policies are crucial for economic recovery and understanding the effects of increased government spending on the economy is of great importance, particularly for policymakers. This information is summarized in terms of multipliers, that quantify the rise in output as a result of a \$1 increase in government spending. Recently, the literature has made great strides in going beyond the average effects of government spending, and distinguishing the impact based on the state of the underlying economy. Empirical research on the state-dependence of the fiscal multiplier is an active area of research, with a lack of consensus on the degree of state dependence in the effectiveness of government spending across good and bad times.¹ Moreover, our understanding of the fiscal transmissions at play across different states of the economy is relatively limited.

This paper contributes to both the empirical and theoretical literature on the state-dependence of government spending. While the earlier literature has focused on whether the multipliers are larger when there is slack in the economy, this paper establishes that the interaction of unemployment and inflation matters in determining the government spending multipliers. Specifically, the government spending multiplier is different depending on the nature of recessions, and it is important to distinguish whether they coincide with high or low inflation.

We propose a New Keynesian model featuring downward nominal wage rigidity (DNWR) with two different sources of business cycle fluctuations: demand and supply shocks.² During an expansion, the conventional effects of government spending prevail in the model, with consumption being crowded out due to rising real interest rate driven by higher inflation. The DNWR constraint becomes relevant during a recession, as it prevents nominal wages from decreasing, which result in real wages being higher than the market-clearing level and causing unemployment to rise.

For the labor market equilibrium, real wage, which is nominal wage deflated by price level, matters. In a recession nominal wage are likely to go down regardless of the source of recession. However, the two different sources of fluctuations lead to a price level response in opposing directions, and consequently different labor market outcomes in recessions. In a demand-driven recession, inflation falls and thus DNWR prevents real

¹See, for example, [Auerbach and Gorodnichenko \(2012\)](#), [Nakamura and Steinsson \(2014\)](#), and [Ramey and Zubairy \(2018\)](#).

²The presence of DNWR in the US is well documented, for example, [Card and Hyslop \(1996\)](#); [Kahn \(1997\)](#); [Barattieri et al. \(2014\)](#); [Daly and Hobijn \(2014\)](#); [Fallick et al. \(2020\)](#); [Grigsby et al. \(2019\)](#); [Murray \(2019\)](#); [Jo \(2021\)](#).

wages from falling. As real wages are higher than the optimal level, expansionary fiscal policy leads to a smaller increase in real wages and marginal costs in a recession. Thus, it limits the rise in inflation and subsequently the real interest rate. As a consequence, it results in less crowding-out effects and a larger government spending multiplier than in an expansion. On the other hand, in a supply-driven recession, inflation goes up, and even if the DNWR constraint binds, it has a limited impact on real wages and unemployment. Government spending raises real wages, inflation and real interest rates to a greater degree, crowding out consumption. Thus, expansionary government spending is relatively less effective in stimulating output in a supply-driven recession.

In our quantitative model, we simulate a deep recession to match the trough of the Great Recession and find that the government spending multiplier is 1.7 in a demand-driven recession, and 0.5 in a supply-driven recession and during expansions. This difference arises because DNWR does not bind in a supply-driven recession or in an expansion. We further investigate the nonlinearity of the effects of increased government spending and show that government spending multipliers are larger as the degree of recession deepens. Multipliers in demand-driven recessions are consistently larger or equal to those in supply-driven recessions, as a marginal increase in government spending reduces labor market distortions more effectively in a severe recession. Additionally, we find that government spending multipliers decrease as the size of government spending shocks gets larger, indicating that the effectiveness of additional government spending in stimulating output diminishes as its size increases. Lastly, we show that the main results are robust to the consideration of alternative preferences and different degrees of price and wage rigidity.

Next, we provide empirical evidence to support these findings. We first focus on time series evidence based on historical macroeconomic data for the United States, following [Ramey and Zubairy \(2018\)](#). This long time series data spanning 1889 to 2015 helps us to exploit time variation in government spending, and also allows us to distinguish between periods of high unemployment accompanied with high and low inflation, historically. Consistent with our model predictions, we find evidence that the government spending multiplier is statistically significantly larger in high unemployment periods accompanied with low inflation, classified as a demand-driven recession, than a high unemployment period accompanied with high inflation, i.e. a supply-driven recession. In addition, we document that there is no evidence of a difference in multipliers based on inflation in the low unemployment periods, which also supports DNWR as a potential mechanism that becomes relevant only in bad times. Furthermore, consistent with our theoretical model, we show empirically that multipliers are larger, the deeper the recession is, in the case of

both high and low inflation.

In order to strengthen the connection between our theoretical and empirical results, we simulate our quantitative model and employ local projections to estimate multipliers. We classify the sources of fluctuations in the simulated data using a comparable classification method as employed in our empirical analysis. This simulation exercise reaffirms that government spending is more effective in low-inflation recessions compared to high-inflation recessions or expansions.

We also conduct a regional analysis, exploiting variation in military procurement contracts across U.S. states, for the sample period 1966 to 2018, in the spirit of [Nakamura and Steinsson \(2014\)](#) to provide further empirical support for our theoretical findings. The U.S. regional-level analysis is useful to show how the effects of government spending vary depending on the relative business cycle conditions after controlling for any aggregate general equilibrium effects. The time fixed effects in the regional analysis account for any changes in the monetary policy regime or taxes and financing responses to changes in government spending. We find that the effects of government spending on the economy are larger during a recession, particularly when it coincides with low inflation. Importantly, this regional approach also enables us to utilize a new dataset quantifying a DNWR measure across U.S. states from [Jo \(2021\)](#) to directly test our proposed mechanism. We find larger effects of government spending in a recession when states face higher levels of DNWR and low inflation—conditions that align with a demand-driven recession in our theoretical framework.

The paper has four contributions. In our theoretical model, we rely on DNWR as a key mechanism. In a model with DNWR, the response of real wages is of primary importance in driving unemployment in the model. We firstly show that the difference in the response of inflation across demand and supply shocks leads to different consequences for real wages even if the nominal wages are downward bound. Thus, real wage frictions transmitted by the joint behavior of nominal wages and inflation are at the heart of exploiting DNWR as a way to generate asymmetries in the business cycle.³

Second, as a consequence of these findings, we show that the source of fluctuation matters for the size of the fiscal multiplier, particularly in a recession. The distinction between recessions and expansions alone might not be sufficient when considering the state-dependent government spending multiplier and the nature of the recession constitutes an important factor. Notably, the same increase in government spending will have a larger output multiplier in a low-inflation recession driven by demand shocks, versus

³See related insights in [Benigno and Ricci \(2011\)](#); [Schmitt-Grohé and Uribe \(2016, 2017\)](#); [Dupraz, Nakamura, and Steinsson \(2019\)](#)

one where the recession is accompanied with high inflation driven by supply shocks.

Some well-established methodologies that have considered state dependence of the government spending multipliers exploiting historical time series data have found limited evidence of larger multipliers across periods of slack in the economy. Our third contribution is to exploit rich historical data to show that these same estimation strategies yield statistically significantly larger multipliers in periods of slack accompanied with low inflation relative to periods with high inflation. This finding aligns with our theoretical framework and helps reconcile disagreements on the relative size of spending multipliers in periods of slack accompanied with low inflation relative to periods with high inflation, depending on the nature of the recessions in the chosen sample periods. We provide further empirical evidence of how in addition to the nature of recession, the depth of the recession also matters for the size of spending multipliers.

Lastly, our cross-sectional analysis using U.S. regional-level data enables us to test the mechanism from our theoretical model directly, as we employ a new dataset quantifying DNWR measures across U.S. states over time. We find that the local spending multiplier is larger in a demand-driven recession with a high degree of DNWR than in a supply-driven recession, which is consistent with our theoretical predictions.

1.1 Related Literature

Our model contributes to the literature on theoretical explanations behind variations in the size of the government spending multipliers based on the state of the economy.⁴ [Shen and Yang \(2018\)](#) show in a New Keynesian model that the government spending multipliers can be higher in a recession than in a boom when DNWR constraint binds in a recession. We build on the insights in this paper and add supply side considerations, which provide new crucial results. [Michaillat \(2014\)](#) generates countercyclical multipliers of government spending in a search and matching model. During high unemployment periods, the rise in public employment increases labor market tightness to a small degree and also has a smaller crowding out effect on private employment. [Albertini et al. \(2020\)](#) consider a model with incomplete markets and nominal rigidity. They are able to generate state-dependent government multipliers as increased spending reduces unemployment

⁴A larger strand of the theoretical literature has considered how the stance of monetary policy affects the government spending multiplier. Notably, they show in a New Keynesian model, the spending multipliers are much larger at the ZLB than in normal times. See, for example, [Christiano et al. \(2011\)](#), [Woodford \(2011\)](#), and [Eggertsson \(2011\)](#). Somewhat related to our work on distinguishing between sources of fluctuations, [Mertens and Ravn \(2014\)](#) find that the size of the fiscal multiplier depends on the type of shock that pushed the economy into the liquidity trap. In particular, they show that when the liquidity trap is due to a non-fundamental shock, supply-side fiscal instruments have a large multiplier, and demand-side fiscal instruments have a small multiplier. The reverse is true when the liquidity trap is caused by a fundamental shock.

and thus unemployment risk and precautionary savings to a greater extent during high unemployment periods. In departure from this literature, we further distinguish based on the nature of a recession and establish that the interactions between unemployment and inflation play a critical role in the magnitude of the multiplier during periods of high unemployment.

[Ghassibe and Zanetti \(2022\)](#) also present a model of differential fiscal multiplier depending upon the source of business cycle fluctuations, featuring search and matching frictions in a goods market. They find that policies stimulating aggregate demand, like increased government spending are more effective in demand-driven recessions, whereas policies affecting aggregate supply have larger multipliers in supply-driven recession. We differ in the main friction of the model - our paper examines the role of DNWR. While [Ghassibe and Zanetti \(2022\)](#) use a real model, we present a model with nominal frictions which predicts comovement between output and inflation, allowing us to identify the sources of the business cycle. We also present empirical findings based on our transmission mechanism for unemployment, inflation, and DNWR. Lastly, they argue that government spending multipliers are larger in a supply-driven expansion compared to a demand-driven expansion, whereas our paper claims that government spending multipliers are the same during expansions, regardless of the source of business cycle fluctuations, both from a theoretical and empirical perspective.

Downward nominal wage rigidity has also been explored as a way to generate asymmetric multipliers, considering differences in response to expansionary versus contractionary government spending. [Barnichon, Debortoli, and Matthes \(2022\)](#) consider a model with incomplete markets and DNWR and generate asymmetric and state-dependent effects of government spending. In a small open economy model with DNWR, [Born, D’Ascanio, Müller, and Pfeifer \(2019\)](#) show that the real exchange rate and output respond asymmetrically to negative and positive government spending shocks under a peg, and support their theoretical results with empirical findings. Our work differs in further emphasizing the role of the source of fluctuations in characterizing the state-dependent government spending multipliers.

Our paper also contributes to the large empirical literature on state-dependent fiscal policy, notably one that explores whether the government spending multiplier differs based on the state of the economy. Some studies like [Auerbach and Gorodnichenko \(2012, 2013\)](#) find distinctly larger spending multipliers in recessions than in expansions. [Ramey and Zubairy \(2018\)](#), on the other hand, do not find multipliers larger than 1 in any state of the economy, and limited evidence of significantly larger multipliers during periods of slack. We extend this analysis to show that conditioning on the interactions between in-

flation and unemployment is important and find evidence of spending multipliers larger than 1 in a low inflation recession, which is statistically significantly larger than the multipliers in expansions and high inflation periods of slack.

Defense contracts have been used to identify the cross-sectional local multipliers by many others, most notably by [Nakamura and Steinsson \(2014\)](#).⁵ The focus of studies employing defense contracts has typically been on quantifying local multipliers, their spillover effects and potentially the mapping to the aggregate multiplier. See [Chodorow-Reich \(2019\)](#) for a survey of this local multiplier literature that exploits state-level variation. [Bernardini, De Schryder, and Peersman \(2020\)](#) use U.S. state level data to study the effects of fiscal policy in the decade surrounding the Great Recession, and find larger multipliers when a state is in a recession or had a high level of household indebtedness. Our paper uses US regional panel data to identify the state-dependent local multipliers depending upon the source of the business cycle. We further exploit the US regional degree of DNWR to show empirical findings consistent with our model results, namely that the local multiplier is higher in a U.S. region with a high degree of DNWR in a low inflation recession.

The remainder of the paper is organized as follows. Section 2 introduces a New Keynesian model with DNWR. We provide the main intuition behind the model in Section 3. Section 4 presents quantitative results that the spending multipliers depend on the state of the economy and the source of fluctuations. Section 5 and Section 6 provide empirical evidence of our proposed model with historical time series data and US state-level annual panel data, respectively. Section 7 concludes.

2 New Keynesian model featuring DNWR

Our baseline model is a New Keynesian model with government spending subject to a DNWR constraint, featuring two sources of the business cycle. We build upon the work of [Shen and Yang \(2018\)](#), who introduced a DNWR constraint into a New Keynesian model with government spending and a preference shock, by incorporating supply-side considerations with a productivity shock. Although DNWR serves as the primary mechanism in our model to generate asymmetry, we depart from previous studies by emphasizing the importance of the source of business cycle fluctuations. The two types of shocks: demand and supply, have different implications for inflation, and we show how inflation can alleviate the impact of DNWR on employment.

⁵[Dupor and Guerrero \(2017\)](#), [Auerbach, Gorodnichenko, and Murphy \(2019\)](#), and [Demyanyk et al. \(2019\)](#), among others.

2.1 Households

A representative household chooses consumption c_t , labor n_t , and nominal bonds B_t to maximize utility over an infinite time horizon:

$$\max_{\{c_t, n_t, B_t\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \prod_{j=0}^t \beta_j \frac{[c_t - \chi(n_t)^\varphi]^{1-\sigma}}{1-\sigma},$$

where β_j is the time varying discount factor in period j , and serves as a demand shock. σ is the inverse of the intertemporal elasticity of substitution and $1/(\varphi - 1)$ is the Frisch elasticity of labor supply. We consider [Greenwood, Hercowitz, and Huffman \(1988\)](#) (GHH) preferences, allowing consumption and labor complementarity and no wealth effect on labor supply.⁶ There are a continuum of consumption goods $c_t(i)$ where $i \in [0, 1]$. The composite consumption is aggregated with the [Dixit and Stiglitz \(1977\)](#) aggregator, $c_t = \left(\int_0^1 c_t(i)^{(\theta-1)/\theta} di \right)^{\theta/(\theta-1)}$. Households are subject to the following period t budget constraint,

$$P_t c_t + B_t + T_t = W_t n_t + R_{t-1} B_{t-1} + \int_0^1 \Gamma(i) di,$$

where P_t is the aggregate price index, B_t is nominal bond, T_t is lump-sum tax, W_t is the nominal wage rate, and R_{t-1} is the nominal interest rate between $t - 1$ and t , and $\Gamma(i)$ is the profit from ownership of firm i .

We assume that nominal wage adjustment is constrained downward, as proposed by [Schmitt-Grohé and Uribe \(2016\)](#).

$$W_t \geq \gamma W_{t-1}, \quad \gamma > 0. \tag{1}$$

The parameter γ governs the degree of DNWR. Nominal wages cannot fall below the previous period's wage when γ is greater than one, while nominal wages are fully flexible when γ is zero. If we assume $\gamma > 0$, nominal wages are not fully flexible and the labor market does not clear all the time. Actual employment used for production (n_t) can be lower than labor supply (n_t^s) when a shock drives the DNWR constraint to bind. Nominal wages and employment must satisfy the complementary slackness condition:

$$(n_t^s - n_t)(W_t - \gamma W_{t-1}) = 0. \tag{2}$$

When the DNWR constraint is not binding ($W_t > \gamma W_{t-1}$), the labor market clears ($n_t^s =$

⁶We adopt the GHH preferences for analytical tractability purposes. We additionally perform quantitative analysis using preferences that allow a wealth effect on labor supply.

n_t) and the unemployment rate is zero. When the DNWR constraint is binding, there is involuntary unemployment ($n_t < n_t^s$), as the households' willingness to work at the prevailing wage is larger than labor demand. We define the unemployment rate as, $u_t = (n_t^s - n_t)/n_t^s \times 100$.

2.2 Firms

The final good y_t is produced with a continuum of intermediate goods, $y_t(i)$, $i \in [0, 1]$, with the technology:

$$y_t = \left(\int_0^1 y_t(i)^{\frac{\theta-1}{\theta}} di \right)^{\frac{\theta}{\theta-1}}.$$

Firms in this market operate under perfectly competitive conditions. Profits are given by

$$P_t y_t - \int_0^1 p_t(i) y_t(i) di.$$

Firms maximize profits subject to the above production technology. The implied demand functions for intermediate goods are $y_t(i) = (p_t(i)/P_t)^{-\theta} y_t$. The aggregate price level is given by $P_t = \left(\int_0^1 p_t(i)^{1-\theta} di \right)^{1/(1-\theta)}$.

Intermediate good i is produced using labor:

$$y_t(i) = A_t n_t(i), \tag{3}$$

where A_t is technology, and serves as an exogenous source of supply disturbances. Given the output level $y_t(i)$ chosen in period t , cost minimization implies marginal cost as given by $mc_t(i) = w_t/A_t$. Following [Calvo \(1983\)](#), a fraction $1 - \omega$ of intermediate firms can optimally choose their prices each period. Firms that get a chance to reset their prices in period t choose their price to maximize the expected sum of discounted future profits. Suppose firm i has the chance to adjust the price in period t , then, $P_t^*(i)$ is chosen so as to maximize

$$\max_{P_t^*(i)} \mathbb{E}_t \sum_{j=0}^{\infty} \omega^j \lambda_{t,t+j} y_{t,t+j}(i) \left[\frac{P_t(i)^*}{P_{t+j}} - mc_{t+j}(i) \right]$$

subject to demand for intermediate goods $y_{t,t+j}(i) = \left(P_t^*(i)/P_{t+j} \right)^{-\theta} y_{t,t+j}$, where $\lambda_{t,t+j} = \mathbb{E}_t \prod_{k=1}^j \beta_{t+k} (\lambda_{t+k}/\lambda_t)$ is the stochastic discount factor for real j -period ahead profit. The full set of optimizing conditions characterizing the equilibrium are shown in the [Appendix A.1](#).

2.3 Closing the model

Monetary policy follows a Taylor rule, and the gross nominal interest rate R_t responds to the deviations of the inflation rate and output from their own steady states, which is summarized as,

$$\frac{R_t}{R} = \left(\frac{\pi_t}{\pi} \right)^{\alpha_\pi} \left(\frac{y_t}{y} \right)^{\alpha_y},$$

where π and y stand for the steady-state level of inflation and output, respectively. The government collects lump-sum taxes T_t to balance the government budget constraint each period:

$$g_t = \frac{T_t}{P_t}.$$

The aggregate market clearing condition is ⁷

$$y_t = c_t + g_t.$$

We assume that the discount factor, aggregate productivity, and government spending shocks follow AR(1) processes:

$$\ln(\beta_t/\beta) = \rho^\beta \ln(\beta_{t-1}/\beta) + \epsilon_t^\beta, \quad (4)$$

$$\ln(A_t/A) = \rho^A \ln(A_{t-1}/A) + \epsilon_t^A, \text{ and} \quad (5)$$

$$\ln(g_t/g) = \rho^g \ln(g_{t-1}/g) + \epsilon_t^g, \quad (6)$$

where $\epsilon_t^\beta \sim \text{iid}N(0, \sigma_\beta^2)$, $\epsilon_t^A \sim \text{iid}N(0, \sigma_A^2)$, and $\epsilon_t^g \sim \text{iid}N(0, \sigma_g^2)$.

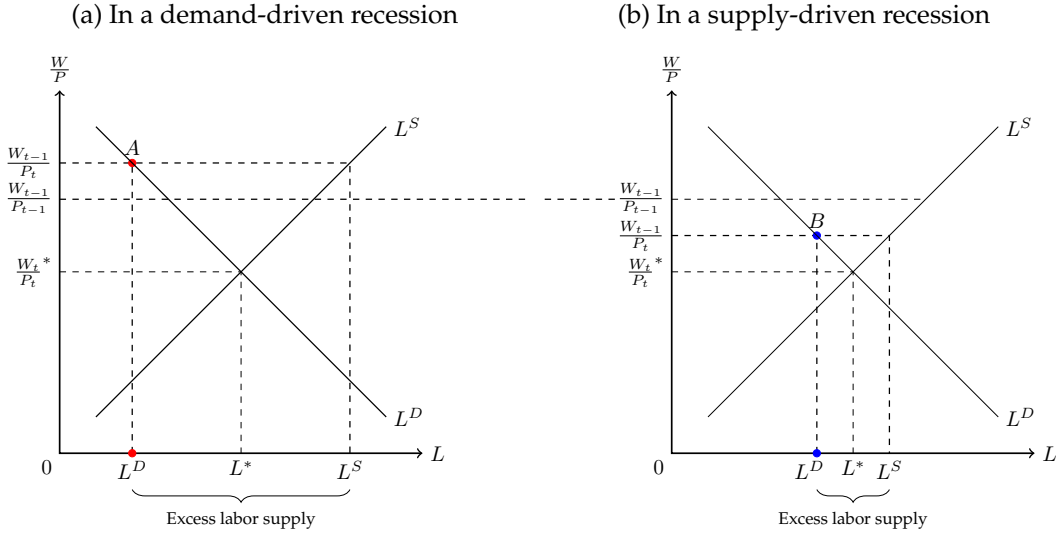
3 Intuition for state-dependent spending multipliers

In this section, we describe the main mechanisms at play in the model. We formally derive analytical results that help illustrate how the government spending multipliers depend on the state of the economy and inflation jointly. For the sake of tractability, we assume that $\gamma = 1$, so nominal wages cannot adjust downward, and that $\alpha_y = 0$ in the monetary policy rule. In this section, we provide the intuition for our analytical results, which can be found in Appendix [A.1](#).

In the model, output and inflation co-move along the demand-driven business cycle,

⁷We have purposely kept the model simple in order to be able to drive analytical results in the next section. However, we can show in the context of our quantitative model that the overall results are unaffected if we consider capital as an input in the production function and thus include investment in the model or if we consider distortionary taxes instead of lump-sum taxes to finance government spending or a monetary policy rule with interest rate smoothing.

Figure 1: Labor market equilibrium in a demand-driven vs. supply-driven recession



Notes: The figure illustrates labor market outcomes in a demand-driven recession (left panel) and a supply-driven recession (right panel). Both recessions result in a market clearing real wage that is lower than the previous real wage. The opposite responses of inflation in both recessions lead to different labor market equilibrium. Points A and B represent the labor market equilibrium in a demand-driven and supply-driven recession, respectively. Both figures have the same level of the previous real wage (W_{t-1}/P_{t-1}) and the current market clearing wage (W_t/P_t)*.

generated by preference shocks whereas output and inflation move in the opposite direction along the supply driven business cycle, generated by technology shocks (see Proposition A.1). A negative preference shock generates a demand-driven expansion, where households prefer to increase consumption in the current period. This increase in goods demand raises labor demand, with an increase in real wages, and thus marginal cost, leading to higher inflation. A positive preference shock, in contrast, leads consumers to postpone their consumption, resulting in a demand-driven recession accompanied with a decline in inflation. On the other hand, a positive productivity shock raises the marginal product of labor, leading to an increase in supply of goods, which generates a supply-driven expansion in output. The reduction in marginal cost lowers inflation. A negative productivity shock causes a supply-driven recession, accompanied with high inflation.

The difference in the inflation response across preference and productivity shock recessions plays a crucial role in determining labor market outcomes when DNWR constraint is present. Figure 1 illustrates the labor market equilibrium in a demand-driven (left panel) and a supply-driven recession (right panel). In a recession, assume that the previous real wage (W_{t-1}/P_{t-1}) is higher than the current market-clearing real wage (W_t/P_t)* due to either a positive discount factor shock (shown in Figure 1a) or a negative productivity shock (shown in Figure 1b). In a recession, nominal wage is not allowed

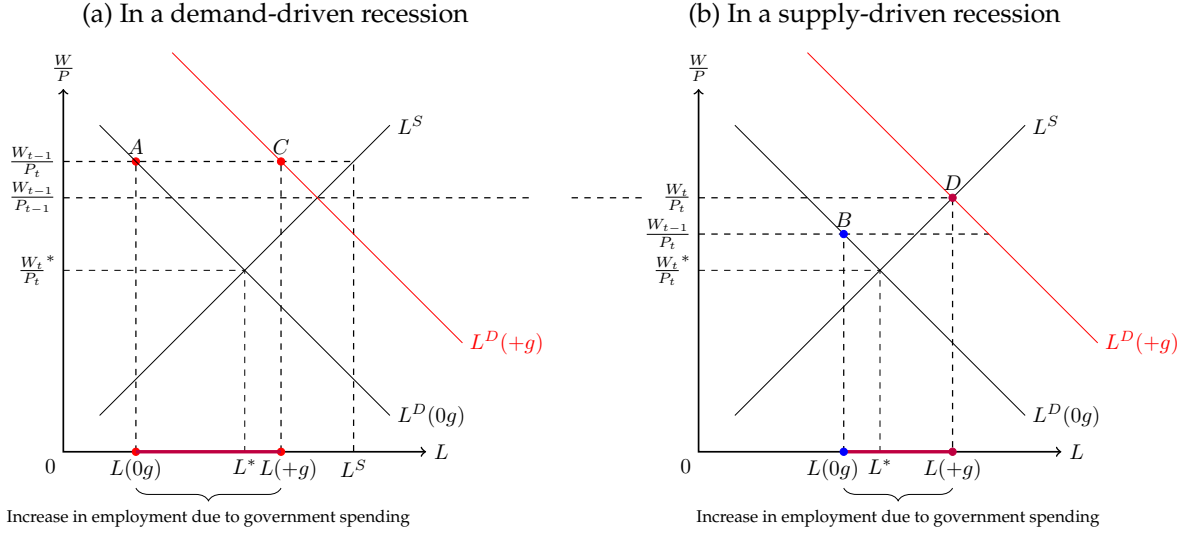
to adjust downward since the DNWR constraint imposes $W_t \geq W_{t-1}$. In contrast, price inflation responds immediately. In response to a contractionary discount factor shock, inflation decreases, which subsequently leads to an increase in the real wage and a reduction in labor demand. Point A in Figure 1a represents the combination of real wage and labor used in production in a demand-driven recession. On the other hand, when there is a negative productivity shock, inflation rises, lowering real wages. Point B in Figure 1b becomes a labor market outcome in a supply-driven recession. For both cases, the quantity of labor demand (L^D) is less than the quantity of labor supply (L^S), leading to an excess labor supply. The equilibrium level of labor is demand-determined in the recession, or $L = L^D = \min\{L^D, L^S\}$ as $L^D \leq L^S$ when DNWR binds. From this example, we can clearly see that inflationary pressure in a supply-driven recession helps adjust real wages downward when the DNWR constraint binds. Consequently, the equilibrium quantity of labor used in production is higher in a supply-driven recession, and there is more excess labor supply or involuntary unemployment in a demand-driven recession.

Now let us introduce a one-time increase in government spending. In the absence of the DNWR constraint, the model reduces to a standard New Keynesian model and is fully symmetric and the government spending multiplier (M_y) is acyclical (see Proposition A.2). However, in the presence of DNWR constraint, government spending multipliers (M_{DNWR}) are larger as long as DNWR is a binding constraint, given the underlying business cycle shocks. When the DNWR constraint binds, an increase in government spending raises equilibrium labor used in production without increasing nominal wage. Additionally, there is also no inflationary pressure on prices with the binding DNWR constraint since real wages, and consequently, marginal costs do not change. The real interest rate thus remains unchanged, ruling out crowding out effects on private consumption.⁸

Consider a government spending increase of a size that is sufficient to relax the DNWR constraint in a supply-driven recession, but not in a demand-driven recession. Consequently, the DNWR constraint still binds in a demand-driven recession but not in a supply-driven recession. Figure 2 illustrates this increase in government spending in both recessions as a rightward shift in labor demand in Figure 1. Since we assume preferences that lead to no wealth effect on labor supply, the labor supply curve does not shift. Note that the size of the shift in labor demand is the same for both cases. In a demand-driven recession, this increase in government spending is not enough to achieve full employment. The new equilibrium is point C in Figure 2a, which shows that the increase in

⁸This statement is true under absolute DNWR. Once we allow degree of DNWR $\gamma < 1$ in the quantitative analysis, the real interest rate rises in response to an increase in government spending. The mechanism at play under absolute DNWR is thus essentially the same as a constant real rate monetary policy in the absence of DNWR, and we can show analytically that it results in the same size multiplier.

Figure 2: An increase in government spending in a demand-driven vs. supply-driven recession



Notes: The figure illustrates the effect of an increase in government spending on labor in a demand-driven recession (left panel) and a supply-driven recession (right panel). Points A and B represent the labor market equilibrium in a demand-driven and supply-driven recession, respectively, without government spending. An increase in government spending shifts labor demand curve to right, resulting an equilibrium point C (left panel) and D (right panel).

government spending can effectively raise labor without raising real wage. In contrast, this increase in labor demand due to expansionary government spending raises equilibrium real wage in a supply-driven recession, moving to equilibrium point D in Figure 2b. The increase in labor in a demand-driven recession (the horizontal distance between A and C in Figure 2a) is higher than the increase in labor in a supply-driven recession (the horizontal distance between B and D in Figure 2a). Therefore, an increase in output caused by an increase in government spending is larger in a demand-driven recession than in a supply-driven recession.

The key takeaway from this analysis is that government spending is more effective as long as the DNWR constraint is binding, and inflation is low. In such situations, government spending can have a larger impact on output, particularly in demand-driven recessions. In a recession, both a positive discount factor shock and a negative productivity shock can lead to the DNWR constraint to bind (see Lemma A.1). Due to the contrasting responses of inflation to these shocks, the size of government spending required to escape the binding DNWR constraint differs across the type of recession. As long as government spending is not enough to relax DNWR constraint, government spending has an effectively larger output multiplier (see Lemma A.2). Given the same sized productivity shock and discount factor shock, the analysis shows that the government spending

shock required to achieve zero unemployment is larger in a demand-driven recession than in a supply-driven recession (see Lemma A.3). In conclusion, we can derive Proposition 1 which states that the government spending multiplier is likely to be larger in a demand-driven recession than in a supply-driven recession for a given level of government spending.

Proposition 1. For a same sized preference and productivity shock,

$$\begin{aligned} & \text{the spending multiplier in a demand-driven recession} \geq \\ & \text{the spending multiplier in a supply-driven recession} \geq \\ & \text{the spending multiplier in an expansion,} \end{aligned}$$

for a given size of government spending shock.

Proof. The proof is available in Appendix A.1.2. □

4 Quantitative analysis of the state-dependent government spending multipliers

In this section, we simulate a calibrated quantitative model to generate the government spending multipliers under various scenarios, distinguishing between expansions and recessions and considering alternative sources of business cycles. To solve a model with an occasionally binding DNWR constraint, we use the OccBin toolkit of Guerrieri and Iacoviello (2015).⁹

4.1 Parameter calibration

The steady-state discount factor (β) is set to be 0.99, implying that the steady-state quarterly real interest rate is 1%. Intertemporal elasticity of substitution, σ , is 1, assuming a log utility function. We set the Frisch elasticity of labor supply ($1/(\varphi - 1)$) to be 0.5, implying φ equals to 3, which is in line with the macro estimates of Frisch elasticity (Chetty et al. (2011)). χ is set to ensure the steady-state level of labor to be 1. The elasticity of substitution across intermediate goods (θ) is set to be 7.67, implying the steady-state price mark-up is 15%. We set $\omega = 0.75$, implying that the firms have on average one chance to reset their price in a year. The coefficients on inflation and output of monetary policy are set at $\alpha_\pi = 1.5$ and $\alpha_y = 0.05$. We set γ , governing the degree of DNWR, as 0.98, which is the lower bound of γ from Schmitt-Grohé and Uribe (2017). This allows at most 8%

⁹Using OccBin, we were able to closely replicate Shen and Yang (2018)'s results with their calibration, where they solve the model with a fully nonlinear solution method. OccBin replicates fully nonlinear solutions when the nonlinear constraints are binding, but it does not capture the expectation effects before the constraint becomes binding.

decline in nominal wages annually, which is a conservative assumption on downward nominal wage rigidity.¹⁰

Following [Fernández-Villaverde et al. \(2015\)](#), we set the persistence of the discount factor shock (ρ_β) to 0.8. The persistence of the productivity shock (ρ_a) is set to be 0.96, in line with [Smets and Wouters \(2007\)](#). We estimate the AR(1) process using HP filtered detrended real government spending data from 1960 to 2019 and set $\rho_g = 0.81$. Standard deviations of the shocks are chosen to match the standard deviations of cyclical component of HP filtered real output and real government spending, and the standard deviation of GDP deflator inflation, leading to $\sigma_\beta = 0.012$; $\sigma_g = 0.285$; $\sigma_a = 0.09$.

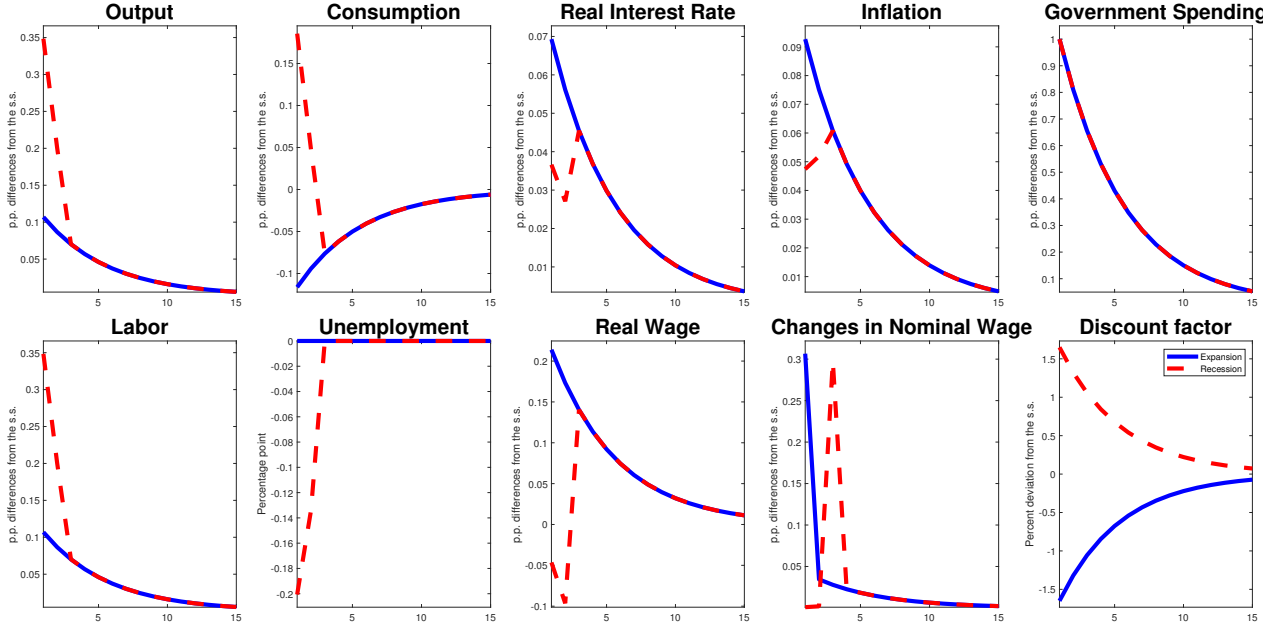
4.2 State-dependent government spending multipliers

We begin by examining how the economy responds to both contractionary and expansionary supply and demand shocks. We choose the size of contractionary supply and demand shock in order to match the average output gap during the Great Recession. According to the Congressional Budget Office estimates, the average output gap from 2008 to 2010 was 4%.

In order to investigate the effects of government spending shocks based on the state of the economy and the source of fluctuation, we first consider a demand driven business cycle. In order to generate a Great Recession sized downturn, we consider $\pm 1.7\%$ deviations from steady-state value of the discount factor shock. This results in an asymmetric demand-driven business cycle. In a demand-driven recession, the binding DNWR constraint as well as deflation further exacerbates labor market outcome, leading to a much deeper recession than the size of expansion. More details of the response of the economy to the underlying business cycle fluctuations are available in [Appendix A.2.1](#). Next, at the same time as the discount factor shock hits the economy, we consider the effects of a government spending shock. [Figure 3](#) plots the differences in impulse responses with and without a government spending shock. Notably, the solid lines depict the differences in impulse responses following an expansionary discount factor shock with and with-

¹⁰Equation (2) implies that when there is an excess labor supply, $n_t^s > n_t$, the DNWR constraint binds, $W_t = \gamma W_{t-1}$. Based on the previous equation, we can calibrate γ using the hourly wage growth rates when we had a large increase in unemployment during the Great Recession. [Schmitt-Grohé and Uribe \(2016\)](#) point out that nominal hourly wage growth rates reflect the long-run growth, while the model abstracts from it. Thus, we deflate the hourly wage growth rate by the long-run growth rate of the United States. The average hourly earnings quarterly growth rate (BLS series ID: CES0500000003) from 2008 to 2010 is 0.6% and the long-run average quarterly growth rate in real GDP from 1947 to 2019 is 0.8%. This implies γ equals 0.998. The recent literature indeed assigns higher values to γ than 0.98. For example, [Auclert and Rognlie \(2020\)](#) impose $\gamma = 1$. [Dupraz, Nakamura, and Steinsson \(2019\)](#) allow annual growth rate as 2.3% with absolute DNWR, resulting in $\gamma = 0.995$. [Barnichon, Debortoli, and Matthes \(2022\)](#) allow annualized wage deflation up to 4%, which corresponds to $\gamma = 0.99$. We consider alternative degrees of price and wage rigidity as robustness checks, in [Appendix A.2.4](#).

Figure 3: Effects of government spending shock for demand-driven business cycle



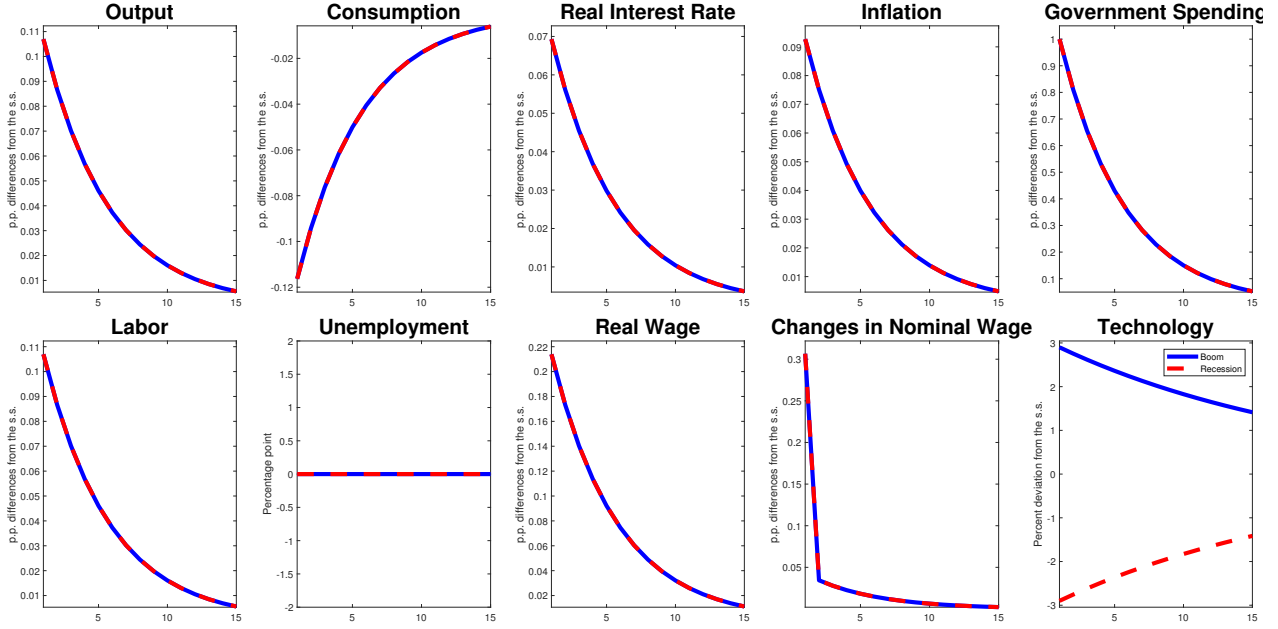
Notes: This graph displays the differences of impulse responses with government spending compared to those without government spending in a demand-driven expansion (solid lines) and a demand-driven recession (dashed lines). A $\pm 1.7\%$ deviation of the discount factor shocks is imposed. All graph is drawn in terms of the percentage points differences from its steady-state except the unemployment rate and the discount factor. The y-axis of the unemployment rate is the percentage point.

out a simultaneous positive government spending shock. Likewise, the dashed line plots differences in impulse responses after a contractionary discount factor shock, with and without a simultaneous positive government spending shock. Consequently, each impulse response represents the changes in macroeconomic variables resulting from a 1% increase in government spending coinciding with a demand-driven business cycle.

An increase in government spending does not immediately raise nominal wages in a demand-driven recession, while it does raise nominal wages in a demand-driven expansion. This is because nominal wages are initially constrained to adjust downward in a recession (See Figure A.4). While the nominal wage is fixed in a recession, inflation increases due to a rise in demand. As a result, the real wage declines, stimulating the labor market. In contrast, during an expansion, the real wage increases immediately in response to heightened demand. This increase in real wage dilutes the effect of the increase in government spending. Therefore, an increase in labor due to expansionary government spending is smaller in a boom compared to a bust. Similarly, the increase in government spending reduces the unemployment rate in a recession when the DNWR is the binding constraint, while it does not affect unemployment in an expansion.

As the discount shock recovers back to the steady state and the effects of expansion-

Figure 4: Effects of government spending shock for supply-driven business cycle



Notes: This graph displays the differences of impulse responses with government spending compared to the one without government spending in a supply-driven expansion (solid line) and a supply-driven recession (dashed line). A $\pm 2.9\%$ deviation of the technology shocks is imposed. All graph is drawn in terms of the percentage points differences from its steady-state except the unemployment rate and the discount factor. The y-axis of the unemployment rate is the percentage point.

any government spending comes in to play, nominal wages begin to adjust upward in a recession. Once there are upward adjustments of nominal wages, which are not subject to any frictions, the DNWR constraint become irrelevant, and the impulse responses during a demand-driven recession coincide with a demand-driven expansion after 4 periods.

Furthermore, the increase in inflation is weaker in a demand-driven recession than in a demand-driven expansion. An increase in government spending in a recession lowers real wage and thus marginal cost. This weakens the response of inflation, caused by an increase in demand, leading to a smaller response of nominal interest rate. Consequently, there is a smaller increase in the real interest rate in a recession, limiting the crowding-out effect on private consumption. To summarize, an increase in government spending is more effective in a demand-driven recession when the DNWR constraint binds because it raises the quantity of labor without a rise in nominal wages, and thus has less inflationary pressure leading to a smaller rise in real interest rates. In conclusion, the effects of government spending shocks is state-dependent in a demand-driven business cycle and much larger in a recession than in an expansion. These results for a demand-driven business cycle are also consistent with [Shen and Yang \(2018\)](#).

Figure 4 displays the differences in impulse response in a supply-driven business cy-

cle with and without government spending. When we consider a Great Recession sized recession, a supply-driven business cycle is symmetric after $\pm 2.9\%$ deviations from the steady-state value of productivity. The drop in aggregate productivity in this case is not enough for the DNWR constraint to bind, leading to symmetric responses of labor. Since the DNWR constraint does not bind in a supply-driven recession (refer to Figure A.5), the responses of the macroeconomic variables to a government spending shock coinciding with the supply shocks are the same regardless of the state of the economy. This also shows that if instead of a recession driven by demand shocks, we consider the same sized recession driven by supply shocks, the main result of Shen and Yang (2018) of state dependent government spending multipliers disappears. We find the same multiplier along the supply-driven business cycle, illustrating the importance of the source of recession for the size of the multiplier.

Table 1: Cumulative output and consumption multipliers by the source of fluctuation

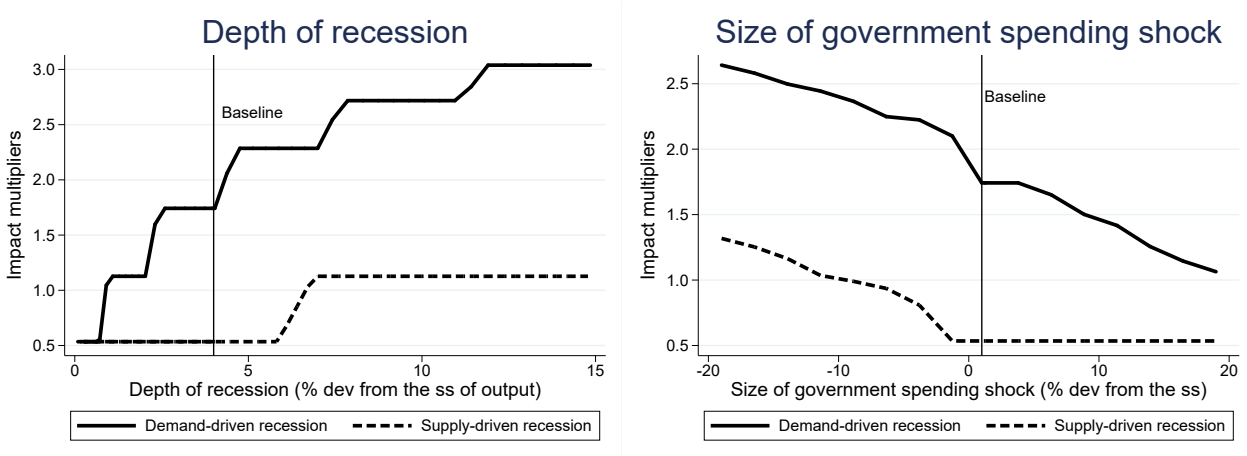
		Demand-driven business cycle			Supply-driven business cycle		
		Impact	4 quarters	20 quarters	Impact	4 quarters	20 quarters
Output Multiplier	Expansion	0.535	0.535	0.535	Expansion	0.535	0.535
	Recession	1.742	1.129	0.879	Recession	0.535	0.535
Consumption Multiplier	Expansion	-0.465	-0.465	-0.465	Expansion	-0.465	-0.465
	Recession	0.742	0.129	-0.121	Recession	-0.465	-0.465

Notes. This table reports the cumulative output and consumption multipliers in an expansion and a recession depending on the source of fluctuation. A $\pm 1.7\%$ discount factor shock is imposed for the demand-driven business cycle, while a $\pm 2.9\%$ technology shock is imposed for a supply-driven business cycle. The cumulative output and consumption multipliers are calculated as $\sum_{i=0}^{k-1} \frac{\Delta y_{t+i}}{(1+r_{t+i})} / \sum_{i=0}^{k-1} \frac{\Delta g_{t+i}}{(1+r_{t+i})}$ and $\sum_{i=0}^{k-1} \frac{\Delta c_{t+i}}{(1+r_{t+i})} / \sum_{i=0}^{k-1} \frac{\Delta g_{t+i}}{(1+r_{t+i})}$, respectively, where Δ denotes the level differences of each variable with and without government spending and r_t is the real interest rate.

Table 1 summarizes the cumulative output and consumption multipliers depending on the state of the economy and the source of fluctuation. The multipliers are state-dependent (countercyclical) in the demand-driven business cycle, taking a value above 1 (1.7 on impact) in a recession and 0.5 in an expansion. This larger multiplier in the recession is driven by a positive multiplier for consumption, at least in the short-run, while the consumption multiplier is negative during an expansion. On the other hand, the multipliers are acyclical in the supply-driven business cycle. The results demonstrate that both the underlying economic states and the source of fluctuations are crucial in determining the size of the spending multipliers.

We also consider alternative preferences a la King, Plosser, and Rebelo (1988), which allow for wealth effects on labor supply in Appendix A.2.2. In this case, DNWR binds in a supply driven recession also, and we show that while the multiplier in a supply driven

Figure 5: Impact multipliers by the depth of recession and the size of government spending shock



Notes: The left figure illustrates the impact multipliers during demand-driven and supply-driven recessions, considering the depth of the recession, in terms of percentage deviation from steady state output, for a given size of government spending shock. The right figure, on the other hand, displays the impact multipliers based on the size of the government spending shock, in terms of percentage deviation of government spending from its steady state, for a given depth of recession.

recession is larger than in an expansion. However, it is still the case that the multiplier in a supply-driven recession is smaller than the multiplier in a demand-driven recession, though the differences across the states are smaller. We also show that our results of larger multipliers in demand driven recessions than supply driven recessions are robust to considering trend inflation (Appendix A.2.3), and alternative degrees of price and wage rigidity (Appendix A.2.4).

4.3 Non-linearity of government spending multipliers

This section demonstrates the non-linearity of government spending multipliers, which depend on the size of business cycle fluctuations (Section 4.3.1), and the size and sign of government spending innovations (Section 4.3.2).

4.3.1 Size of business cycle fluctuations

The magnitude of the spending multipliers for output is contingent on the size of underlying fluctuations for the same-sized government spending shocks. The left panel of Figure 5 plots the impact output multipliers based on the depth of a demand-driven recession (solid line) and supply-driven recession (dashed line). The vertical line represents our baseline calibration, where we match the depth of the Great Recession in terms of output deviation from steady state. The demand-driven recession multipliers are always greater than or equal to the supply-driven recession multipliers, which confirms our Proposition

1. Ultimately, the size of the multiplier depends on the extent of the binding DNWR.

In a severe demand-driven recession, an additional increase in demand from a spending shock significantly raises output, leading to a considerably larger output and consumption multiplier. When the recessions are mild and do not trigger DNWR to bind, both demand and supply recession multipliers are the same as the expansion multipliers. As the demand-driven recession deepens, it leads to the DNWR condition binding for a longer period, resulting in larger multipliers of the magnitude of 2.5 to 3. Each step in the panel indicates the number of periods the DNWR constraint is binding. Similarly, supply-driven recession multipliers become larger than the expansion multipliers as the supply-driven recession becomes more severe, as DNWR starts binding in this case as well. Also, as we consider deep recessions with significant deviation of output from steady state, the multipliers in both types of recessions are larger than in expansions, but the multipliers in a supply-driven recession remain lower than the demand-driven recession multipliers.

4.3.2 Size and sign of government spending

In addition to the size of business cycle fluctuations, the size and sign of government spending also play a role in determining the magnitude of spending multipliers. The right panel of Figure 5 illustrates the impact multipliers based on the size and sign of government spending, given the same underlying business cycle fluctuation, with multipliers in a demand-driven recession (solid line), and in a supply-driven recession (dashed line). The vertical line represents our baseline case, where we have a 1% increase in government spending shock. The results are again consistent with Proposition 1, where demand-driven recession multipliers are larger than supply-driven recession multipliers for a given size of business cycle and spending shock. As the size of the government spending shock increases, the impact multipliers decrease during a demand driven recession.¹¹ This is because marginal increases in government spending reduce labor market distortion to a smaller extent, which becomes less effective in leading to a sizable increase in output. However, during a supply driven recession, the size of the government spending shock does not affect the spending multiplier, provided it is a positive spending shock, since DNWR does not bind.

In a demand-driven recession, a negative government spending intervention of 1% exacerbates the labor market distortion, raising the spending multiplier to above 2. The multipliers become even larger when we consider even bigger negative spending shocks. On the other hand, since DNWR constraint does not bind in a supply-driven recession,

¹¹For a large enough government spending shock, i.e. larger than 25% deviation from steady state, the multiplier in a demand driven recession becomes smaller than 1, and eventually converges to the expansion multiplier, indicating that a large enough fiscal shock can make the DNWR constraint irrelevant.

the sign of government spending intervention (a 1% increase or decrease) does not affect the size of the multipliers. However, when we have a large enough negative government spending shock to trigger the DNWR constraint binding, government spending multipliers in a supply-driven recession start to rise, as a negative government spending shock can be regarded as a negative demand shock.

Empirically, [Barnichon, Debortoli, and Matthes \(2022\)](#) provide evidence that negative shocks to government spending result in larger multipliers than positive shocks. Recently, [Ben Zeev et al. \(2023\)](#) show that these differences based on the sign of the shock are not statistically significantly different. While our model predictions are not about the unconditional size of the multiplier based on the sign of the shock alone, our results are partly consistent with the additional findings in [Barnichon, Debortoli, and Matthes \(2022\)](#), which suggest that the spending multiplier for a negative shock to government spending is larger in a slack state than the multiplier for a positive shock. We show that this is likely to be true in a demand driven recession, and only unconditionally in any recession or period of slack if the size of the spending shock is large enough. Our results suggest that fiscal austerity is particularly harmful in a demand-driven recession.

5 Time series empirical evidence

In the previous section, we theoretically establish that government spending multipliers in a recession can differ based on the inflation response, driven by different source of recessions. In order to empirically investigate whether the government spending multipliers are state-dependent and if the nature of recession matters, we exploit historical time series data for the US. Additionally, we examine the nonlinearity of government spending multipliers, particularly whether the depth of recession matters for the size of multipliers, as predicted by our theory. Lastly, we demonstrate that our theoretical framework is consistent with the empirical results by comparing multipliers estimated with simulated data from the quantitative model and empirical estimates.

5.1 Econometric Methodology

We employ a one-step IV estimation procedure for estimating fiscal multipliers as introduced in [Ramey and Zubairy \(2018\)](#), using state-dependent local projections a la [Jordà \(2005\)](#). In departure from this earlier analysis, we distinguish not only between low and high unemployment periods but also consider the interaction between unemployment and inflation. We separately consider high unemployment periods accompanied by low inflation (i.e., inflation below a threshold), which can be considered as periods where contractionary demand shocks are a major source of fluctuations. When high unemployment

coincides with high inflation (i.e., inflation above the threshold), we consider contractionary supply shocks as a dominant source of fluctuations.

We consider the following state-dependent local projection model,

$$\sum_{j=0}^h y_{t+j} = \sum_d \mathbb{I}(\text{State } d) \left[\gamma_{d,h} + \phi_{d,h}(L) z_{t-1} + m_{d,h} \sum_{j=0}^h g_{t+j} \right] + \omega_{t+h},$$

where y_t is real GDP and g_t is real government spending, both normalized by trend GDP.¹² The normalization and consideration of cumulative GDP and government spending variables ensure that the coefficient m_h can be interpreted as the cumulative government spending multiplier at horizon h in a given state. The control variables, z_t , consists of lagged values of the military news variable, GDP and government spending. We include 4 lags of each variable and we allow all coefficients to change based on the state of the underlying state of the economy.

In our baseline specification, we consider an indicator function for the *pre-existing* state of the economy when the shock hits, which corresponds to $\mathbb{I}(L(u_{t-1}))$, the state where lagged unemployment is low, $\mathbb{I}(H(u_{t-1})) \times \mathbb{I}(L(\pi_{t-1}))$, periods of lagged high unemployment and low inflation and $\mathbb{I}(H(u_{t-1})) \times \mathbb{I}(H(\pi_{t-1}))$, periods of lagged high unemployment and high inflation. We use $\mathbb{I}_{t-1} \times shock_t$ as the instruments for the respective interaction of cumulative government spending with the state indicator, where we consider the narrative military news variable from [Ramey and Zubairy \(2018\)](#) as our baseline shock. Since this LP-IV approach allows us to consider multiple instruments, in a later specification, we also consider the case with both military news and identification based on [Blanchard and Perotti \(2002\)](#).

Our data set constitutes of quarterly data for the U.S. spanning 1889Q1-2015Q4 from [Ramey and Zubairy \(2018\)](#), including GDP, unemployment rate, government spending and GDP deflator. This historical data provides large variation in government spending, the unemployment rate and periods of high and low inflation. We define inflation as year-over-year growth of the GDP deflator. In order to define states, we consider high or low unemployment periods where the unemployment rate is above or below the threshold of 6.5%, respectively, as considered by [Ramey and Zubairy \(2018\)](#). We consider high and low inflation periods as being defined as above or below a threshold of 4%, which corresponds to the 75th percentile of inflation for our full sample. Using this distinction in the inflation rates to distinguish between the type of recession implies that the Great

¹²The trend or potential GDP is constructed by using a sixth-order polynomial. This normalization for GDP and government spending ensures we do not have to use the average share of government spending to GDP to convert government spending into GDP units to get the multipliers. [Ramey and Zubairy \(2018\)](#) show that this approach can bias the multipliers, particularly in samples where there is large variation in spending as a share of GDP. We test robustness to an alternative definition of potential GDP in Table A.4.

Depression, for the most part, and the Great Recession were demand-driven recessions. On the other hand, recessions in the mid 1970s and early 1980s start off as supply-driven recessions before negative demand forces take over.¹³

5.2 Evidence of State Dependent Multipliers: Type of Recession

Table 2 presents our baseline results for state-dependent multipliers, where we consider the military news variable to identify the government spending shocks. The multipliers are defined as the cumulative multipliers, which account for the cumulative dynamics of output and government spending, as advocated for in Mountford and Uhlig (2009). The linear and the two-state multipliers based on the level of unemployment rate replicate the findings of Ramey and Zubairy (2018) (Columns 1, 2 and 5, 6). Notably, the spending multipliers are not statistically different across periods of high and low unemployment. Similarly, the multipliers are not statistically different from each other depending on the level of inflation (Columns 3 and 7).

When we consider the three different states of the economy, we find a 2 year integral multiplier of 0.6 in the low unemployment state, which is close to the linear multiplier. At the 2 year horizon, the multiplier in the high unemployment state significantly differs based on inflation: it is close to 1 in the low inflation state and statistically zero in the high inflation state. The multiplier in the high unemployment and low inflation state is statistically significantly larger than the multiplier in the low unemployment state. At the 4 year horizon, the low unemployment multiplier is close to the linear multiplier, at 0.7. Once again, the multipliers are statistically significantly different when the unemployment rate is high, based on the state of inflation. They are estimated to be about 0.8 and 0.2 across the low and high inflation states, respectively.¹⁴ This provides evidence consistent with our theoretical findings: the government spending multiplier is significantly larger in a demand-driven recession than a supply-driven recession.¹⁵

¹³The classification of these different states along with data on military news, unemployment rate and inflation are shown in Figure A.6 in Appendix A.3. The classifications of the recessions, particularly in the 1970s and 1980s are consistent with the ones presented in Blanchard and Quah (1989). For example, the early 1980s recession started with supply-driven forces as the 1978-1979 Iranian revolution and 1980-1981 Iran-Iraq war led to a cut in oil production, resulting in high inflation. The Fed tightening monetary policy resulted in a demand-driven recession for the latter part of the early 1980s recession.

¹⁴Table 2 reports 2 and 4 year horizon multipliers only, but Figure A.7 in Appendix A.3 shows the multipliers and corresponding standard error bands over the entire 5 year horizon.

¹⁵The empirical estimates for the average multiplier in a demand-driven recession are close to 1 and smaller than the multipliers from the quantitative model. This is essentially because we simulate a deep recession in the model and as further explorations of the quantitative model reveal, the multiplier in a demand-driven recession can even be less than one in a mild recession. Table 2 also reports that the multiplier in a supply-driven recession (periods with high unemployment accompanied with high inflation) is smaller than the multiplier in an expansion. At first glance, it seems contradictory to Proposition 1, but those results are derived under the assumption of the equal sized business cycle for a given size and sign

Table 2: State-dependent fiscal multipliers for output: military news shocks

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	2-year integral output				4-year integral output			
Σg_t	0.66 (0.07)				0.71 (0.04)			
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))$		0.59 (0.09)		0.59 (0.09)		0.67 (0.12)		0.67 (0.12)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))$		0.60 (0.09)				0.68 (0.05)		
$\Sigma g_t \times \mathbb{I}(L(\pi_{t-1}))$			0.73 (0.11)				0.68 (0.07)	
$\Sigma g_t \times \mathbb{I}(H(\pi_{t-1}))$			0.60 (0.06)				0.70 (0.05)	
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.99 (0.19)				0.78 (0.06)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				-0.19 (0.15)				0.20 (0.10)
P-value from the test								
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))$		0.95				0.92		
$\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(\pi_{t-1}))$			0.32				0.82	
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.09				0.47
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.00				0.00
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.00				0.00
Effective first-stage F statistics								
Linear	19.38				11.22			
$\mathbb{I}(L(u_{t-1}))$		8.44		8.07		10.85		10.56
$\mathbb{I}(H(u_{t-1}))$		403.28				130.20		
$\mathbb{I}(L(\pi_{t-1}))$			9.72				5.08	
$\mathbb{I}(H(\pi_{t-1}))$			73.18				45.86	
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				56.37				249.55
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				45.89				71.63
Observations	493	493	493	493	485	485	485	485

Notes: The top panel reports the 2 and 4 year cumulative multiplier along with associated heteroskedasticity and autocorrelation robust standard errors below. $\mathbb{I}(L(u_{t-1}))$ and $\mathbb{I}(H(u_{t-1}))$ indicate the state where lagged unemployment is low and high, respectively. $\mathbb{I}(L(\pi_{t-1}))$ and $\mathbb{I}(H(\pi_{t-1}))$ represent the state where lagged inflation is low and high, respectively. The second panel shows p-values testing whether the multipliers are statistically significantly different across states. The last panel reports the [Olea and Pflueger \(2013\)](#) effective first-stage F statistics for military news as an instrument at 2 and 4 year horizons for the relevant subsample. We use the threshold for the 5 percent critical value for testing the null hypothesis that the TLS bias exceeds 10 percent of the OLS bias. For one instrument, this threshold is always 23.1.

Table 3: State-dependent fiscal multipliers for output: both military news and [Blanchard and Perotti \(2002\)](#) as instruments

	(1) 2-year integral output	(2)	(3) 4-year integral output	(4)
Σg_t	0.42 (0.10)		0.56 (0.08)	
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$		0.31 (0.18)		0.36 (0.19)
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$		0.47 (0.04)		0.54 (0.09)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$		0.68 (0.21)		0.79 (0.06)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$		0.22 (0.13)		0.44 (0.09)
P-value from the test				
$\mathbb{I}(L(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(L(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$		0.36		0.32
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$		0.09		0.00
Effective first-stage F statistics				
Linear	36.70 [13.19]		14.44 [15.46]	
$\mathbb{I}(L(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$		42.35 [17.58]		33.86 [18.39]
$\mathbb{I}(L(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$		38.88 [9.25]		16.76 [8.52]
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$		54.12 [10.20]		32.35 [19.50]
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$		77.89 [7.38]		56.94 [8.56]
Observations	493	493	485	485

Notes: The top panel reports the 2 and 4 year cumulative multiplier along with associated heteroskedasticity and autocorrelation robust standard errors below. $\mathbb{I}(L(u_{t-1}))$ and $\mathbb{I}(H(u_{t-1}))$ indicate the state where lagged unemployment is low and high, respectively. $\mathbb{I}(L(\pi_{t-1}))$ and $\mathbb{I}(H(\pi_{t-1}))$ represent the state where lagged inflation is low and high, respectively. The second panel shows p-values testing whether the multipliers are statistically significantly different across states. The last panel reports the [Olea and Pflueger \(2013\)](#) effective first-stage F statistics for military news and [Blanchard and Perotti \(2002\)](#) shocks jointly as instruments at 2 and 4 year horizons for the relevant subsample. The numbers in brackets provide the 5 percent critical value used to test the null hypothesis that the TSLS bias exceeds 10 percent of the OLS bias, which is specification dependent.

The primary reason for conducting a three state analysis and not distinguishing between the inflation rate across the non-slack states is the low instrumental relevance of military news for the low unemployment state corresponding with low inflation. As shown in the bottom panel of Table 2, the instrumental relevance of military news is small in the low unemployment state overall, even without any distinction based on inflation.¹⁶ On the other hand, the alternative leading identification scheme of Blanchard and Perotti (2002), based on assuming that government spending does not respond to contemporaneous output and macroeconomic variables in the same quarter, has very low instrumental relevance in the high unemployment state.¹⁷

In order to address these instrumental relevance issues across various states and horizons, we conduct the same analysis using both military news and Blanchard and Perotti (2002) shocks as joint instruments, shown in Table 3. In this case, the multipliers are overall estimated to be smaller. Column (2) shows that the difference in multipliers across slack states is a function of inflation. The 2-year cumulative multiplier in a low-inflation recession is statistically different and larger than the multiplier in a high inflation recession. This result aligns with our theoretical predictions that the multiplier in a demand-driven recession is larger than the multiplier in a supply-driven recession. The gap between the multipliers in the high unemployment state across low and high inflation states becomes even more statistically significant at the 4 year horizon (Column (4)).

In addition, Columns (2) and (4) report that the multipliers in expansions or low unemployment periods, regardless of inflation, are similar to each other. While it is not possible to completely rule out other explanations for state-dependence of multipliers, the fact that inflation does not affect multipliers in the low unemployment state provides supportive evidence for DNWR, which is at play only in periods of high unemployment.

We further explore the response of additional macroeconomic variables, such as inflation, unemployment, and real interest rate, to government spending shocks in various states of the economy in Table A.12 in Appendix A.3. The findings provide additional

of government spending. Nonetheless, large government spending in a mild supply-driven recession can generate a relatively smaller multiplier, or a large negative government spending in an expansion can lead to a larger multiplier.

¹⁶The tables report the Olea and Pflueger (2013) effective first-stage F-statistics and corresponding critical values, which account for autocorrelation in the error terms, which is inherent in the Jorda method. This test only applies to models with a single endogenous regressor, and works with both one and two instrument case in Tables 2 and 3, since we consider regime subsamples. In models with a single endogenous regressor, the test-statistic and critical values are equivalent to Olea and Pflueger (2013), but in models with two instruments, it prefers a more conservative bound on the bias. A more generalized version of this test a la Lewis and Mertens (2022) works in the case of additional endogenous regressors.

¹⁷Ramey and Zubairy (2018) have already shown that beyond the impact, the instrumental relevance of Blanchard and Perotti (2002) shocks become smaller at longer horizons of 2 and 4 years out, based on the underlying identification assumption.

support for our DNWR mechanism. In line with the theoretical model, we observe that unemployment declines in a demand-driven recession, while it exhibits an insignificant response in supply-driven recessions. This difference across the two types of recessions is statistically significantly different at horizons after 8 quarters. We also examine the response of year-over-year GDP deflator inflation rate, and find that inflation rises in both types of recessions, but the difference is not statistically significantly different across the two states.¹⁸ With a smaller sample from 1915 onward, we also consider the response of the real rate (constructed as a difference between the T-bill rate and inflation) to a government spending shock. Consistent with the predictions of the theoretical model, we find that real rate rises more in supply driven recessions than demand driven recessions, and this difference is statistically significantly different at relevant horizons.¹⁹

One could argue that the employment rate is more likely to be subject to a time-varying threshold capturing the natural rate of unemployment. We conduct additional robustness checks with alternative thresholds for both inflation and unemployment rate in Appendix A.3. Table A.6 uses time-varying thresholds for both the unemployment rate and inflation. Table A.7 and A.8 estimate the multipliers using time-variant thresholds for inflation with military news shocks and both news and Blanchard and Perotti (2002) shocks as instruments, respectively. Overall, the results are robust. We find that the multipliers in a low inflation recession are larger than high inflation recessions. The results are also robust to considering a lower threshold for inflation, at 3%, in order to define high and low inflation periods (Table A.5).

Another concern could be that taxes or generally the method of financing affects the propagation of government spending shocks across the various states of the economy. We add lagged values of tax revenues as a share of output as additional controls, interacted with our state of the economy indicator functions in our baseline estimation equations, and find that the results are unchanged. See Table A.9 in Appendix A.3.

We have also conducted the historical time series analysis after controlling for real interest rates, which are available from 1915 onward. We find that the aggregate multipliers

¹⁸In order to assess whether monetary policy could be playing a role in accommodating the fiscal expansions, we consider the response of the T-bill rate (data for which is available from 1915 onward, so a slightly shorter sample period) to a government spending shock. The T-bill rate response is not statistically significantly difference across demand and supply driven recessions, which seems to rule out a difference in monetary policy response to spending innovations across the two types of recessions.

¹⁹Another possible way to validate the model predictions is to estimate the differential responses of aggregate wages in response to government spending shocks depending on the source of the business cycle. However, this is hard to implement. Firstly, there is no such long time series for wages available. Aggregate wages also suffer from composition bias (Solon, Barsky, and Parker (1994)), as there are changes in the composition of workers in recessions, and aggregate wages are thus constructed with a relatively larger weight on highly-paid workers.

Table 4: Nonlinearity of government spending multipliers: depth of recession

	(1) 2-year integral output	(2) 4-year integral output
$\Sigma g_t \times \mathbb{I}(\text{Deep low-inf recession})$	0.99 (0.21)	0.78 (0.05)
$\Sigma g_t \times \mathbb{I}(\text{Med low-inf recession})$	0.43 (0.26)	0.25 (0.27)
$\Sigma g_t \times \mathbb{I}(\text{Med high-inf recession})$	-0.06 (0.22)	-0.76 (0.43)
$\Sigma g_t \times \mathbb{I}(\text{Deep high-inf recession})$	0.46 (0.26)	1.12 (0.30)
P-value from the test		
$\mathbb{I}(\text{Deep low-inf recession}) = \mathbb{I}(\text{Med low-inf recession})$	0.10	0.05
$\mathbb{I}(\text{Deep high-inf recession}) = \mathbb{I}(\text{Med high-inf recession})$	0.07	0.00
$\mathbb{I}(\text{Deep low-inf recession}) = \mathbb{I}(\text{Deep high-inf recession})$	0.13	0.27
Effective first-stage F statistics		
$\mathbb{I}(\text{Deep low-inf recession})$	82.58	168.81
$\mathbb{I}(\text{Med low-inf recession})$	432.62	305.06
$\mathbb{I}(\text{Med high-inf recession})$	20.00	18.83
$\mathbb{I}(\text{Deep high-inf recession})$	29.84	16.40
Observations	493	485

Note: The table presents 2-year and 4-year cumulative multipliers and robust standard errors for demand-driven and supply-driven recessions. Demand-driven recessions have low inflation (<4%), while supply-driven recessions have high inflation (>4%). Deep and medium recessions correspond to unemployment rates >7% and 5-7%, respectively. The table also shows p-values for multiplier differences across states and [Olea and Pflueger \(2013\)](#) effective first-stage F-statistic for military news as instrument in the relevant subsample. We use the threshold for the 5 percent critical value for testing the null hypothesis that the TSLS bias exceeds 10 percent of the OLS bias. For one instrument, this threshold is always 23.1.

across the different states of the economy are only slightly different with and without this additional control and are still overall larger in low inflation periods of slack than high inflation periods coinciding with high unemployment. See [Table A.10](#) and [Table A.11](#) in [Appendix A.3](#). Lastly, we also show the robustness of our results for an alternative definition of potential GDP, see [Table A.4](#).

5.3 Nonlinear Government Spending Multipliers: Depth of Recession

The size of a recession plays a crucial role in determining spending multipliers. As our theory in [Section 4.3](#) demonstrates, DNWR is likely to bind for a longer period in deeper recessions. Consequently, this leads to larger multipliers in deeper recessions compared to milder ones. We test this hypothesis using our time series framework by distinguishing between mild and deep recessions based on different unemployment thresholds. Specif-

ically, we examine multipliers in deep and mild demand-driven and supply-driven recessions separately. A deep recession is defined as one where the unemployment rate is above a threshold of 7 percent, and a mild recession is one where the unemployment rate is between 5 and 7 percent. We differentiate the nature of recession based on the inflation rate, with demand-driven (supply-driven) recession corresponding to periods with inflation rates below (above) 4 percent.

Table 4 shows that the multiplier is larger in a deep low inflation recession than a mild low inflation recession. This is also true in high-inflation recessions, with larger multipliers observed in deep recessions. The second panel shows that these differences are also statistically significantly different. Overall, these results align with our theoretical predictions and provide additional suggestive evidence that DNWR serves as an important transmission mechanism for government spending shocks in recessions. These empirical results are also consistent with the findings of [Caggiano et al. \(2015\)](#) who find evidence of nonlinearity and larger multipliers only in deeper recessions, and not an average recession relative to expansions, using a different sample period and alternative estimation methodology.

5.4 Comparison of Time Series Results and Model Simulations

In our empirical analysis, in contrast to our quantitative structural model, we rely on the unemployment rate and inflation to classify demand-driven and supply-driven recessions. To further strengthen the link between our empirical findings and the theoretical model, we compare the estimated multipliers employing simulated data from the quantitative model with their empirical counterparts.

We first generate a sample of 504 observations under random shock processes of ϵ_β , ϵ_A , and ϵ_g for 1000 iterations. For each iteration, we use the simulated unemployment rate and inflation to classify regimes and estimate state-dependent multipliers using local projections. Specifically, low unemployment and low inflation states are identified as periods when the unemployment levels or inflation fall below their respective median. We use innovations in government spending, ϵ_t^g , as our instrumental variable.

Table 5 compares cumulative multipliers from our empirical analysis in the earlier section and the estimates from simulated data from our quantitative model. Column (1) reports 2-year cumulative multipliers, where we consider 3 states and rely on military news shock. Column (2) reports 2-year cumulative multiplier where we consider 4 distinct states of the economy and use both military news shocks and [Blanchard and Perotti \(2002\)](#) identified shocks. The numbers in parentheses represent standard errors. Column (3) and (4) report the 2-year median of the cumulative multiplier across 1000 simulations

Table 5: Two-year cumulative multipliers: time series data vs. simulated data

	(1)	(2)	(3)	(4)
	2-year integral output			
	Data		Simulation	
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))$	0.5949 (0.0905)		0.7276 [0.1312]	
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$		0.3075 (0.1826)		0.7950 [0.2201]
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$		0.4664 (0.0416)		0.6699 [0.1720]
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$	0.9886 (0.1878)	0.6805 (0.2092)	1.1359 [0.2501]	1.1359 [0.2501]
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$	-0.1920 (0.1503)	0.2234 (0.1315)	0.6070 [0.2103]	0.6070 [0.2103]
IV	Military News	Military News BP shocks	ϵ_t^g	ϵ_t^g
Observation	504	504	504	504
Iteration			1000	1000

Notes: Column (1) corresponds to Column (4) of Table 2 and Column (2) corresponds to Column (4) of Table 3. Last two columns report the median estimates of multipliers using the simulated data of a sample size 504 across 1000 iterations. The number in brackets indicates the standard deviations of the estimated multipliers.

of the quantitative model. The number in brackets indicate the standard deviations of the estimated multipliers.

Multipliers based on simulated data exhibit a larger multiplier in a low-inflation recession than in a high-inflation recession. Additionally, multipliers from simulated data are similar across different types of expansions and smaller than ones in a low-inflation recession. These results align well with our empirical findings. Furthermore, the multipliers based on simulated data in a low-inflation recession are very close to the empirical estimates identified relying on military news shocks, even though it is not a targeted moment. These comparisons suggest that our theoretical findings are consistent with empirical findings, thereby enhancing the link between our theoretical and empirical results.²⁰

²⁰We find that empirical estimates of multipliers in high-inflation recessions are considerably lower than the simulated estimates. This discrepancy likely arises because shocks to the business cycle and government spending are symmetrically distributed in simulated data, while the magnitude of government spending remains relatively large during supply-driven recessions compared to demand-driven recessions in historical data.

6 US regional empirical evidence

In order to directly explore whether DNWR plays a key mechanism in determining state-dependent fiscal spending multipliers, we exploit U.S. regional-level variation in output, inflation, and military procurement spending associated with aggregate military buildups. This approach allows us to employ rich regional-level data on the degree of binding DNWR, made possible due to recent advances using individual-level panel data on wages. While national-level military spending does not exhibit large variations except during wartime, regional-level military spending data allows us to exploit variation across both time and cross-sectional dimensions. In contrast with the time series analysis in the previous section, which might potentially suffer from confounding factors such as aggregate monetary policy regimes, U.S. regional panel regression with time-fixed effects enables us to control for these aggregate factors, like the stance of monetary policy.²¹

These U.S. regional-level regressions yield estimates of local multipliers, which are not exactly the same as the aggregate multipliers discussed in previous sections. However, [Chodorow-Reich \(2019\)](#) argues, based on theoretical explorations, that the typical empirical cross-sectional multiplier seems to provide a lower bound for an aggregate multiplier, corresponding to a closed economy, deficit-financed multiplier in a no-monetary policy response regime.

6.1 US regional-level data

For our regional analysis, we use U.S. states as the geographical unit. The regional-level annual data sample spans from 1969 to 2018. Regional-level nominal GDP is sourced from the U.S. Bureau of Economic Analysis (BEA). We use regional-level inflation and military spending data from [Nakamura and Steinsson \(2014\)](#), which runs until 2008, and extend it to 2018. The details are provided in the Appendix [A.4](#).

We measure the extent of binding DNWR as the difference between the share of workers whose year-over-year hourly wage growth rates are (i) zero and (ii) negative for each U.S. state and year from 1979 to 2018, constructed by [Jo \(2021\)](#) using the Current Population Survey. [Jo \(2021\)](#) shows that in a recession when employment declines, the share of workers with zero wage changes increases disproportionately more than the share of workers with wage cuts.²² [Jo \(2021\)](#) characterizes cyclical properties of nominal wage change distribution using the heterogeneous agent model with various wage rigidity as-

²¹Earlier literature (see [Nakamura and Steinsson \(2014\)](#); [Leeper, Traum, and Walker \(2017\)](#); [Zubairy \(2014\)](#) among others) points out that the government spending multipliers are sensitive to aggregate monetary policy, for example, passive or active monetary policy regimes.

²²[Grigsby, Hurst, and Yildirmaz \(2019\)](#) also show similar US-state-level results from 2008 to 2016.

assumptions commonly used in the literature. [Jo \(2021\)](#) concludes that the model with DNWR is the most consistent with empirical findings: generating excessive countercyclicality of the share of workers having zero wage changes compared to wage cuts. We use this finding to quantify the degree of binding DNWR for each U.S. region across time, providing a valuable tool to examine the relationship between DNWR and fiscal spending multipliers at the regional level.

6.2 Econometric approach

The baseline regression equation for the regional-level analysis is as follows.

$$\frac{Y_{it} - Y_{it-s}}{Y_{it-s}} = \alpha_i + \gamma_t + \sum_d \beta_d \frac{G_{it} - G_{it-s}}{Y_{it-s}} \mathbb{I}(\text{State } d) + \text{Controls} + \epsilon_{it}. \quad (7)$$

where Y_{it} denotes per capita real output in region i in year t and G_{it} denotes per capita real military procurement spending in region i in year t , region-fixed effect, α_i , controls for region-specific trends and time-fixed effect, γ_t , controls for aggregate conditions that are common across regions such as long-run trend inflation or aggregate monetary policy.

We conduct a regional panel regression, regressing two-year differences in per capita output on the two-year differences in per capita military procurement spending. Both variables are normalized by the two-year lagged per capita output. This normalization helps us control for heteroskedasticity across regions, following previous research. We interpret the parameter β_d of interest as an open economy relative spending multiplier depending of the state of economy. Our military spending data is recorded in the fiscal year, whereas all other data is reported in the calendar year. A biannual regression potentially resolves these time differences as they overlap for most of the time period, following [Nakamura and Steinsson \(2014\)](#) and [Dupor and Guerrero \(2017\)](#). We define the state of economy indicators based on regional data on employment, inflation and DNWR. Additional details on how we define indicators representing the state of the economy are available in [Appendix A.4](#).

In order to address endogeneity concerns that the regional-level military spending possibly respond to the current macroeconomic status of each region, we instrument our dependent variables with two variables. The first instrumental variable is the sensitivity of each region's changes in military spending with respect to changes in national military spending, which is introduced by [Nakamura and Steinsson \(2014\)](#). The identifying assumption is that the sensitivity is time-invariant and national military spending is exogenous to relative business cycle conditions of each specific region. We use each region's predicted value of military spending, computed as the estimated elasticity ($\widehat{\psi}_i$) times na-

tional military spending growth $((G_t - G_{t-s})/Y_{t-s})$, as our instrumental variable.²³

We also use Bartik type regional-specific time-varying instrument variable widely used in the previous literature. We construct it as $B_{it} = s_{it}(G_t - G_{t-s})/Y_{t-s}$, where s_{it} is the average level of per capita military procurement spending in that region relative to per capita region output from the previous two years. Using the predetermined share of military spending, we can avoid the reverse causality concern that region differential military spending can be affected by its region-specific current business cycle conditions.

To meet the lead-lag exogeneity condition suggested by [Stock and Watson \(2018\)](#), we include one biannual lag of the growth rate of output, military spending, and both instrumental variables as controls in our analysis.²⁴

6.3 US regional-level estimation results

Table 6 shows the effect of regional-level military procurement spending on output depending upon regional-level employment, inflation, and DNWR. To correct for endogeneity bias, we use both instruments - sensitivity and Bartik instruments interacted with regime indicators. The first two columns of Table 6 show the baseline spending multipliers for the entire sample period. The relative multiplier is 1.7, which lies between the estimates using sensitivity IV and Bartik IV from [Nakamura and Steinsson \(2014\)](#).²⁵ After controlling for lagged variables, as shown in Column (2), the spending multiplier is higher than the one without lagged control variables, while it results in lower instrumental relevance.²⁶ The spending multiplier for a slack period is larger than the one for non-slack period, although these two coefficients are not statistically different from

²³The region-specific sensitivity ψ_i is estimated from the regression equation: $(G_{it} - G_{it-s})/Y_{it-s} = \phi_t + \psi_i(G_t - G_{t-s})/Y_{t-s} + \epsilon_{it}$.

²⁴[Chen \(2019\)](#) and [Ramey \(2020\)](#) point out that instrumental variables can be serially correlated in the US-region-level analysis, not satisfying the lead-lag exogeneity requirement that the external instruments should be uncorrelated with past and future shocks. Thus, adding lagged variables in our regression estimation helps to ensure that our instrumental variables have no serial correlation.

²⁵Note that the sample period in [Nakamura and Steinsson \(2014\)](#) ends in 2008, whereas our sample runs until 2018.

²⁶This is because autocorrelation coefficients on the control variables change over time. As mentioned in the previous section, in the presence of multiple endogenous regressors, [Lewis and Mertens \(2022\)](#) provide the appropriate statistic for a test of weak instrument robust to heteroskedasticity and autocorrelation, particularly in our case where we conduct a single bias test, when testing equal coefficients across regimes. Thus, the table reports the [Lewis and Mertens \(2022\)](#) F-statistics and corresponding critical value, which is application specific, for each case for a bias threshold of 0.1 and a significance level of 0.05. We compute the statistics and critical values using the MATLAB code provided by [Lewis and Mertens \(2022\)](#). The state-level cross-sectional regressions, shown in Tables 6 and 7, are known in the literature to be plagued by weak instrument issues, and in departure from most existing studies we add lagged variables and use the right test statistic, which accounts for autocorrelation and heteroskedasticity. Thus, we encounter an even more severe issue of weak instruments, which suggests a need for caution when interpreting the p-value of difference in coefficients across states. For comparison, we also report the more commonly reported, Kleibergen-Paap rk Wald F statistic in the tables too.

each other (Column (3) and (4)). The spending multipliers differ based on the level of the regional-level inflation (Column (5) and (6)). Both estimates show a larger spending multiplier for the period of high inflation.

The spending multiplier is larger when the US-region-year record high DNWR (Column (7) and (8)), and statistically significant only in the case of high DNWR. In our theoretical framework, binding DNWR serves as a sufficient statistic for *marginal* government spending multipliers when inflation is low. Consistent with theory, we report larger regional multipliers for regions with a relatively high degree of binding DNWR, although it is not statistically different from regional multipliers for regions with a low degree of binding DNWR. It is important to note that the degree of DNWR is not a sufficient statistic for determining *average* government spending multipliers. For instance, the left panel of Figure 5, from the quantitative model, shows that the spending multipliers vary depending on the depth of the recession. Although DNWR becomes a binding constraint, the multipliers in supply-driven recessions are still significantly smaller compared to those in demand-driven recessions of the same depth. This demonstrates that an indicator with high DNWR, which encompasses situations with both high and low inflation and various recession magnitudes, is not an adequate statistic for determining average spending multipliers. This highlights the importance of considering the nature of the recession as well as times when DNWR constraint binds.

The results shown in Table 7 provide clearer evidence of state-dependent spending multipliers when we account for the interaction between employment, inflation, and DNWR. By allowing for these interactions, we can better identify the source of the recession, whether it is driven by demand or supply shocks. Due to lack of individual wage data for the early sample period, we first explore the size of the local spending multiplier based upon employment and inflation for the entire sample period. The spending multiplier is the highest when both employment and inflation are low, periods in which DNWR is most likely to bind (Column (1)). This is in line with our theory that spending is more effective in a demand-driven recession (i.e. low employment and low inflation) than in a supply-driven recession (i.e. low employment and high inflation).

In order to directly examine the role of DNWR as a key mechanism, we analyze data on the degree of DNWR in US regions from 1979 to 2018. We find that the spending multiplier is largest for those US-region-year where a low employment period coincides with a high degree of DNWR (Column (2)). This finding supports our theory that government spending is more effective in stimulating output in a recession when DNWR acts as a binding constraint.

The specification in Column (3) presents four distinct states, allowing us to investi-

gate differential impact on the spending multipliers relying upon employment, inflation, and DNWR. It shows that the estimated effects of spending are largest during periods characterized by high DNWR, low employment and low inflation (i.e. a demand-driven recession). This estimate is statistically different from the estimates for a high DNWR, low employment, and high inflation (i.e. a supply-driven recession) period. In fact, the multiplier in the high DNWR, low employment and low inflation state is the only estimate statistically significantly different from zero. The coefficient in a high DNWR, low employment and high inflation state is not precisely estimated due to a small sample size, and may possibly reflect theoretical predictions that DNWR are not likely to bind in a high inflation recession. Overall, the results are in agreement with our theory that government spending is more effective in a low-inflation recession with binding DNWR than in a high-inflation recession.

7 Conclusion

We study the effectiveness of government spending in stimulating output depending on the source of the business cycle and the state of the economy. We first build a New Keynesian model with DNWR, featuring two different sources of fluctuations: demand and supply shocks. The spending multipliers are different based on the nature of the recession. The simultaneous movement of nominal wage and price matters for the DNWR constraint to have real consequences for labor.

During recessions, nominal wages decline, regardless of the sources of fluctuations. Inflation decreases in a demand-driven recession and increases in a supply-driven recession. As a result, in a demand-driven recession, when nominal wages are constrained from adjusting downwards, the drop in inflation further prevents real wages from decreasing, leading to higher unemployment. Consequently, government spending in this state of the economy becomes more effective, as it increases labor without raising wages and curtails the growth of inflation and the real interest rate, resulting in less crowding out of private consumption. In contrast, during a supply-driven recession, prices adjust upwards while nominal wages face DNWR, leading to limited allocational consequences for labor. This leads to smaller effects of increased government spending in overcoming the labor market distortion. Consequently, the government spending multiplier in a demand-driven recession is larger than in a supply-driven recession.

We provide empirical evidence that corroborates these theoretical results using US historical time series data and US regional panel data. Consistent with theory, we find that the spending multipliers are statistically significantly larger in a low inflation recession than in a high inflation recession. We also provide evidence, in line with the theoretical

model, that the depth of the recession also matters and multipliers are larger for deeper recessions. Furthermore, we provide direct evidence of DNWR as a transmission mechanism and show that the spending multiplier is larger in a US-region with a high degree of DNWR in a demand-driven recession.

Our results provide important implications for the design of economic policy. Notably, we provide evidence that whenever government spending is being considered as a stabilization or a stimulative tool, it is important to recognize the underlying forces driving the economy.

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Table 6: Regional state-dependent spending multipliers: conditioning on two states based on employment, inflation, and DNWR

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
				Output				
$\frac{G_{it}-G_{it-2}}{Y_{it-2}}$	1.67 (0.40)	2.29 (0.92)						
$\frac{G_{it}-G_{it-2}}{Y_{it-2}}\mathbb{I}(H(e_{it}))$			1.38 (0.48)	2.13 (1.35)				
$\frac{G_{it}-G_{it-2}}{Y_{it-2}}\mathbb{I}(L(e_{it}))$			2.33 (0.49)	2.66 (0.81)				
$\frac{G_{it}-G_{it-2}}{Y_{it-2}}\mathbb{I}(H(\pi_{it}))$					2.92 (0.83)	4.09 (1.31)		
$\frac{G_{it}-G_{it-2}}{Y_{it-2}}\mathbb{I}(L(\pi_{it}))$					2.68 (0.98)	1.61 (1.13)		
$\frac{G_{it}-G_{it-2}}{Y_{it-2}}\mathbb{I}(H(DNWR_{it}))$							4.08 (1.16)	2.43 (1.41)
$\frac{G_{it}-G_{it-2}}{Y_{it-2}}\mathbb{I}(L(DNWR_{it}))$							3.81 (1.06)	1.47 (1.26)
P-value from the test								
$\mathbb{I}(H(e_{it})) = \mathbb{I}(L(e_{it}))$			0.11	0.65				
$\mathbb{I}(H(\pi_{it})) = \mathbb{I}(L(\pi_{it}))$					0.71	0.02		
$\mathbb{I}(H(DNWR_{it})) = \mathbb{I}(L(DNWR_{it}))$							0.74	0.28
Observations	2,450	2,350	2,450	2,350	2,354	2,242	1,450	1,150
Period	1966-2018	1966-2018	1966-2018	1966-2018	1969-2018	1969-2018	1979-2018	1979-2018
Controls		Lagged variables		Lagged variables		Lagged variables		Lagged variables
Kleibergen-Paap rk Wald F statistic	258.66	6.96	159.00	2.87	19.53	6.71	16.74	30.51
Lewis and Mertens (2022) F statistic	40.67	9.69	17.74	6.35	11.51	5.57	12.82	10.54
Lewis and Mertens (2022) Critical Value	21.48	21.64	24.98	25.43	24.60	24.09	24.01	25.69
J Statistic	0.17	0.25	4.39	3.56	3.81	4.05	1.29	0.78
Jstat P value	0.68	0.62	0.11	0.17	0.15	0.13	0.52	0.68

Notes: The top panel reports the 2 year regional-level multiplier along with associated regional-level clustered robust standard errors below in parenthesis. $\mathbb{I}(H(e_{it}))$ and $\mathbb{I}(L(e_{it}))$ indicate high and low employment US-region-years, $\mathbb{I}(H(\pi_{it}))$ and $\mathbb{I}(L(\pi_{it}))$ represent high and low inflation US-region-years, and $\mathbb{I}(H(DNWR_{it}))$ and $\mathbb{I}(L(DNWR_{it}))$ denote high and low binding DNWR US-region-years. The second panel shows p-values testing whether the multipliers are statistically significantly different across states. Two instrumental variables are used for the estimation - sensitivity and Bartik instruments. The lagged variables are added as the control variables in Column (2), (4), (6), and (8). In order to assess instrumental relevance, Kleibergen-Paap rk Wald F statistic and [Lewis and Mertens \(2022\)](#) test statistics and the corresponding critical values are reported. J statistics from the overidentification test and the corresponding p-values are reported at the bottom of the table. The overidentification test tests that all instrumental variables are exogenous is not rejected.

Table 7: Regional state dependent spending multipliers: conditioning on three states based on employment, inflation, and DNWR

	(1)	(2) Output	(3)
$\frac{G_{it}-G_{it-2}}{Y_{it-2}}\mathbb{I}(H(e_{it}))$	1.72 (1.13)	0.85 (1.84)	0.76 (1.82)
$\frac{G_{it}-G_{it-2}}{Y_{it-2}}\mathbb{I}(L(e_{it}))\mathbb{I}(L(\pi_{it}))$	3.86 (1.19)		
$\frac{G_{it}-G_{it-2}}{Y_{it-2}}\mathbb{I}(L(e_{it}))\mathbb{I}(H(\pi_{it}))$	-0.16 (1.17)		
$\frac{G_{it}-G_{it-2}}{Y_{it-2}}\mathbb{I}(H(DNWR_{it}))\mathbb{I}(L(e_{it}))$		3.01 (1.41)	
$\frac{G_{it}-G_{it-2}}{Y_{it-2}}\mathbb{I}(L(DNWR_{it}))\mathbb{I}(L(e_{it}))$		1.74 (1.01)	1.54 (0.95)
$\frac{G_{it}-G_{it-2}}{Y_{it-2}}\mathbb{I}(H(DNWR_{it}))\mathbb{I}(L(e_{it}))\mathbb{I}(H(\pi_{it}))$			-12.46 (8.44)
$\frac{G_{it}-G_{it-2}}{Y_{it-2}}\mathbb{I}(H(DNWR_{it}))\mathbb{I}(L(e_{it}))\mathbb{I}(L(\pi_{it}))$			3.12 (1.42)
P-value from the test			
$\mathbb{I}(H(e_{it})) = \mathbb{I}(L(e_{it}))\mathbb{I}(L(\pi_{it}))$	0.16		
$\mathbb{I}(L(e_{it}))\mathbb{I}(L(\pi_{it})) = \mathbb{I}(L(e_{it}))\mathbb{I}(H(\pi_{it}))$	0.05		
$\mathbb{I}(H(e_{it})) = \mathbb{I}(H(DNWR_{it}))\mathbb{I}(L(e_{it}))$		0.25	
$\mathbb{I}(H(DNWR_{it}))\mathbb{I}(L(e_{it})) = \mathbb{I}(L(DNWR_{it}))\mathbb{I}(L(e_{it}))$		0.28	
$\mathbb{I}(H(e_{it})) = \mathbb{I}(H(DNWR_{it}))\mathbb{I}(L(e_{it}))\mathbb{I}(L(\pi_{it}))$			0.22
$\mathbb{I}(H(DNWR_{it}))\mathbb{I}(L(e_{it}))\mathbb{I}(H(\pi_{it})) = \mathbb{I}(H(DNWR_{it}))\mathbb{I}(L(e_{it}))\mathbb{I}(L(\pi_{it}))$			0.06
Observations	2,242	1,150	1,112
Period	1969-2018	1979-2018	1979-2018
Controls	Lagged variables	Lagged variables	Lagged variables
Kleibergen-Paap rk Wald F statistic	3.52	14.57	15.51
Lewis and Mertens (2022) F statistic	6.14	7.87	8.15
Lewis and Mertens (2022) Critical Value	27.53	26.37	26.10
J Statistic	1.89	1.88	3.38
Jstat P value	0.59	0.60	0.50

Notes: The top panel reports the 2 year regional-level multiplier along with associated regional-level clustered robust standard errors below in parenthesis. $\mathbb{I}(H(e_{it}))$ and $\mathbb{I}(L(e_{it}))$ indicate high and low employment US-region-years, $\mathbb{I}(H(\pi_{it}))$ and $\mathbb{I}(L(\pi_{it}))$ represent high and low inflation US-region-years, and $\mathbb{I}(H(DNWR_{it}))$ and $\mathbb{I}(L(DNWR_{it}))$ denote high and low binding DNWR US-region-years. The second panel shows p-values testing whether the multipliers are statistically significantly different across states. Two instrumental variables are used for the estimation - sensitivity and Bartik instruments. The lagged variables are added as the control variables. In order to assess instrumental relevance, Kleibergen-Paap rk Wald F statistic and [Lewis and Mertens \(2022\)](#) test statistics and the corresponding critical values are reported. J statistics from the overidentification test and the corresponding p-values are reported at the bottom of the table. The overidentification test tests that all instrumental variables are exogenous is not rejected.

Online Appendix to “State dependent government spending multipliers: Downward Nominal Wage Rigidity and Sources of Business Cycle Fluctuations”

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A.1 Appendix: Analytics of state-dependent government spending multipliers

An equilibrium is a set of stochastic processes $\{\lambda_t, c_t, w_t, mc_t, R_t, \pi_t, x_t^1, x_t^2, y_t, n_t, n_t^s, u_t, s_t, p_t^*\}_{t=0}^\infty$ satisfying:

$$\lambda_t = (c_t - \chi n_t^\varphi)^{-\sigma} \quad (\text{A.1})$$

$$\chi \varphi n_t^{s\varphi-1} = w_t \quad (\text{A.2})$$

$$\lambda_t = R_t \mathbb{E}_t \frac{\beta_{t+1} \lambda_{t+1}}{\pi_{t+1}} \quad (\text{A.3})$$

$$W_t \geq \gamma W_{t-1}; w_t \geq \gamma \frac{w_{t-1}}{\pi_t} \quad (\text{A.4})$$

$$(n_t^s - n_t)(w_t - \gamma \frac{w_{t-1}}{\pi_t}) = 0 \quad (\text{A.5})$$

When DNWR does not bind ($w_t > \gamma \frac{w_{t-1}}{\pi_t}$), full employment is achieved, $n_t^s = n_t$ and $u_t = 0$. As opposed, if DNWR binds, that is, $w_t = \gamma \frac{w_{t-1}}{\pi_t}$, there is an excess supply of labor, $n_t^s > n_t$ and $u_t > 0$.

$$u_t = \frac{n_t^s - n_t}{n_t^s} \quad (\text{A.6})$$

$$p_t^* = \frac{\theta}{\theta - 1} \frac{x_t^1}{x_t^2} \quad (\text{A.7})$$

$$x_t^1 = \lambda_t y_t mc_t + \omega \mathbb{E}_t \beta_{t+1} \pi_{t+1}^\theta x_{t+1}^1 \quad (\text{A.8})$$

$$x_t^2 = \lambda_t y_t + \omega \mathbb{E}_t \beta_{t+1} \pi_{t+1}^{\theta-1} x_{t+1}^2 \quad (\text{A.9})$$

$$mc_t = \frac{w_t}{A_t} \quad (\text{A.10})$$

$$\pi_t = \left[\frac{1}{\omega} - \frac{1-\omega}{\omega} p_t^{*1-\theta} \right]^{\frac{1}{\theta-1}} \quad (\text{A.11})$$

$$y_t = A_t n_t / s_t \quad (\text{A.12})$$

$$y_t = c_t + g_t \quad (\text{A.13})$$

$$s_t = (1 - \omega) p_t^{*- \theta} + \omega \pi_t^\theta s_{t-1} \quad (\text{A.14})$$

$$\frac{R_t}{R} = \left(\frac{\pi_t}{\pi} \right)^{\alpha_\pi} \left(\frac{y_t}{y} \right)^{\alpha_y} \quad (\text{A.15})$$

, given exogenous stochastic processes $\{g_t, \beta_t, A_t\}_{t=0}^\infty$, which are following AR(1) processes specified as below:

$$\ln \frac{g_t}{g} = \rho^g \ln \frac{g_{t-1}}{g} + \epsilon_t^g \quad (\text{A.16})$$

$$\ln \frac{\beta_t}{\beta} = \rho^\beta \ln \frac{\beta_{t-1}}{\beta} + \epsilon_t^\beta \quad (\text{A.17})$$

$$\ln \frac{A_t}{A} = \rho^A \ln \frac{A_{t-1}}{A} + \epsilon_t^A \quad (\text{A.18})$$

A.1.1 Derivation of IS-PC curves

We derive the IS and the Phillips curve (PC) summarizing equilibrium conditions, (A.1) ~ (A.15). To derive the IS equation, log-linearize both the monetary policy rule (A.15) and the household's intertemporal optimization equation (A.3). Combining the previous two equations yields

$$\widehat{\lambda}_t = \mathbb{E}_t \widehat{\lambda}_{t+1} + \alpha_\pi \widehat{\pi}_t - \mathbb{E}_t \widehat{\pi}_{t+1} + \mathbb{E}_t \widehat{\beta}_{t+1}. \quad (\text{A.19})$$

, where hat variables stand for log-deviations from the steady state and the variable without time subscript represents its steady-state value. Find $\widehat{\lambda}_t$ by log-linearizing the marginal utility of consumption (A.1),

$$\widehat{\lambda}_t = -\frac{\sigma c}{c - \chi n^\varphi} \widehat{c}_t + \frac{\sigma \chi \varphi n^\varphi}{c - \chi n^\varphi} \widehat{n}_t. \quad (\text{A.20})$$

Now let's find the steady-state values of variables. From the production function (A.12), we know that the steady state level of output $y=A$. Note that the steady-state value of s is

zero under the zero inflation steady-state (Galí (2008)). By the market clearing condition (A.13), we find the steady-state consumption is then $c = y - g$. Define the steady-state government spending-to-output ratio as $s_g \equiv \frac{g}{y}$. Then, $c = (1 - s_g)A$. Assume the steady-state labor n equals to labor supply, n^s , which equals to 1. Using Equation (A.2) and (A.10), solve for the model-implied parameter χ assuring $n = 1$ as

$$\chi = \frac{w}{\varphi} = \frac{1}{\varphi} \times A \times mc = \frac{A}{\varphi} \frac{\theta - 1}{\theta}.$$

Substituting the steady-state values to the Equation (A.20) yields

$$\widehat{\lambda}_t = -\frac{\theta(1-s_g)}{\Psi}\widehat{c}_t + \frac{(\theta-1)}{\Psi}\widehat{n}_t, \quad (\text{A.21})$$

where $\Psi = \frac{\theta\varphi(1-s_g)-(\theta-1)}{\sigma\varphi}$. The log linearization of the market clearing condition (A.13) and the production function (A.12) leads

$$\widehat{c}_t = \frac{1}{1-s_g}\widehat{y}_t - \frac{s_g}{1-s_g}\widehat{g}_t \quad (\text{A.22})$$

$$\widehat{y}_t = \widehat{a}_t + (\widehat{n}_t - \widehat{s}_t). \quad (\text{A.23})$$

Galí (2008) shows that $\widehat{s}_t$ equals to zero up to a first-order approximation. Combining (A.19), (A.21), (A.22), and (A.23) yields the NKIS equation:

$$\widehat{y}_t = \mathbb{E}_t \widehat{y}_{t+1} - (\theta-1)(\widehat{a}_t - \mathbb{E}_t \widehat{a}_{t+1}) + \theta s_g(\widehat{g}_t - \mathbb{E}_t \widehat{g}_{t+1}) - \Psi(\alpha_\pi \widehat{\pi}_t - \mathbb{E}_t \widehat{\pi}_{t+1}) - \Psi \mathbb{E}_t \widehat{\beta}_{t+1} \quad (\text{A.24})$$

where $\Psi = \frac{\theta\varphi(1-s_g)-\theta+1}{\sigma\varphi}$.

Now let's derive Phillips curve (PC). The PC can be written in two ways, depending upon whether DNWR binds or not. The first-order approximation of Equation (A.7) and (A.11) yields

$$\widehat{\pi}_t = \frac{(1-\omega)(1-\omega\beta)}{\omega} \widehat{mc}_t + \beta \mathbb{E}_t \widehat{\pi}_{t+1}, \quad (\text{A.25})$$

where $\widehat{mc}_t$ takes two forms. When DNWR does not bind, full employment is achieved ($\widehat{n}_t = \widehat{n}_t^s$). Log-linearization of the Equation (A.2) under the full employment regime yields $\widehat{w}_t = (\varphi-1)\widehat{n}_t$. From the Equation (A.10), we know that $\widehat{mc}_t = \widehat{w}_t - \widehat{a}_t$. Combining previous two equations with Equation (A.23) leads

$$\widehat{mc}_t = (\varphi-1)\widehat{y}_t - \varphi\widehat{a}_t. \quad (\text{A.26})$$

Substituting (A.26) into (A.25) yields the PC curve under the full employment equilibrium:

$$\hat{\pi}_t = \Delta(\varphi - 1)\hat{y}_t - \Delta\varphi\hat{a}_t + \beta\mathbb{E}_t\hat{\pi}_{t+1}, \quad (\text{A.27})$$

where $\Delta = \frac{(1-\omega)(1-\omega\beta)}{\omega}$. When DNWR binds ($\gamma = 1$), we can re-write $\hat{w}_t = \hat{w}_{t-1} - \hat{\pi}_t$. Then,

$$\widehat{mc}_t = \hat{w}_{t-1} - \hat{\pi}_t - \hat{a}_t. \quad (\text{A.28})$$

Substituting (A.28) into (A.25) yields the modified PC curve under the binding DNWR

$$(1 + \Delta)\hat{\pi}_t = \Delta[\hat{w}_{t-1} - \hat{a}_t] + \beta\mathbb{E}_t\hat{\pi}_{t+1}, \quad (\text{A.29})$$

or

$$\hat{\pi}_t = \frac{(1-\omega)(1-\omega\beta)}{\omega + (1-\omega)(1-\omega\beta)}[\hat{w}_{t-1} - \hat{a}_t] + \frac{\omega\beta}{\omega + (1-\omega)(1-\omega\beta)}\mathbb{E}_t\hat{\pi}_{t+1}. \quad (\text{A.30})$$

A.1.2 Analytical results

Now let's consider the two sources of business cycle fluctuations - a preference shock ($\hat{\beta}_{t+1}$) and a productivity shock ($\hat{a}_t$). For tractability, we assume complete DNWR, with $\gamma = 1$. Additionally, nominal interest rates respond to deviations in the inflation rate from its steady state but not to output deviations ($\alpha_\pi > 0$ and $\alpha_y = 0$).

Assumption 1. The sequences of the preference shock ($\mathbb{E}_t\hat{\beta}_{t+1} = b_L < 0$ and $\mathbb{E}_t\hat{\beta}_{t+2} = 0$) and ($\mathbb{E}_t\hat{\beta}_{t+1} = b_H > 0$ and $\mathbb{E}_t\hat{\beta}_{t+2} = 0$) cause a demand-driven expansion and recession, respectively, in period t . The sequence of the technology shock ($\hat{a}_t = a_H > 0$, $\mathbb{E}_t\hat{a}_{t+1} = \rho_a a_H$, and $\mathbb{E}_t\hat{a}_{t+2} = a_L$) and ($\hat{a}_t = a_L < 0$, $\mathbb{E}_t\hat{a}_{t+1} = \rho_a a_L$, and $\mathbb{E}_t\hat{a}_{t+2} = a_H$) drive a supply-driven expansion and recession, respectively, in period t .

Proposition A.1. In response to a preference shock, output ($\hat{y}_t$) and inflation ($\hat{\pi}_t$) co-move, and in response to a technology shock, output and inflation move in the opposite direction. That is,

$$\frac{\partial \hat{y}_t}{\partial \hat{\beta}_{t+1}} < 0; \frac{\partial \hat{\pi}_t}{\partial \hat{\beta}_{t+1}} < 0, \text{ and } \frac{\partial \hat{y}_t}{\partial \hat{a}_t} > 0; \frac{\partial \hat{\pi}_t}{\partial \hat{a}_t} < 0.$$

Proof. Let's consider two independent shock processes. The demand-driven business cycles follow ($\mathbb{E}_t\hat{\beta}_{t+1} = \hat{\beta}_{t+1}$, and $\mathbb{E}_t\hat{\beta}_{t+2} = 0$) where $\mathbb{E}_t\hat{\beta}_{t+1}$ is β_H in a demand-driven recession and $\mathbb{E}_t\hat{\beta}_{t+1}$ is β_L in a demand shock-boom. The supply-driven business cycles are to follow ($\hat{a}_t = \hat{a}_t$, $\mathbb{E}_t\hat{a}_{t+1} = \rho_a \hat{a}_t$, and $\mathbb{E}_t\hat{a}_{t+2} = \hat{a}_{t+2}$) where $(\hat{a}_t, \hat{a}_{t+2}) = (a_H, a_L)$ in a supply-driven boom and $(\hat{a}_t, \hat{a}_{t+2}) = (a_L, a_H)$ in a supply-driven recession. Suppose that the market clearing solution takes the form:

$$\hat{y}_t = A_y \hat{g}_t + B_y \mathbb{E}_t \hat{\beta}_{t+1} + C_y \hat{a}_t + D_y \mathbb{E}_t \hat{a}_{t+1} = A_y \hat{g}_t + B_y \mathbb{E}_t \hat{\beta}_{t+1} + C_y \hat{a}_t + \rho_a D_y \hat{a}_t$$

$$\hat{\pi}_t = A_\pi \hat{g}_t + B_\pi \mathbb{E}_t \hat{\beta}_{t+1} + C_\pi \hat{a}_t + D_\pi \mathbb{E}_t \hat{a}_{t+1} = A_\pi \hat{g}_t + B_\pi \mathbb{E}_t \hat{\beta}_{t+1} + C_\pi \hat{a}_t + \rho_a D_\pi \hat{a}_t.$$

Given the assumptions on shock processes and government spending, the expected output and inflation are

$$\mathbb{E}_t \hat{y}_{t+1} = A_y \mathbb{E}_t \hat{g}_{t+1} + B_y \mathbb{E}_t \hat{\beta}_{t+2} + C_y \mathbb{E}_t \hat{a}_{t+1} + D_y \mathbb{E}_t \hat{a}_{t+2} = \rho_a C_y \hat{a}_t + D_y \hat{a}_{t+2}$$

$$\mathbb{E}_t \hat{\pi}_{t+1} = A_\pi \mathbb{E}_t \hat{g}_{t+1} + B_\pi \mathbb{E}_t \hat{\beta}_{t+2} + C_\pi \mathbb{E}_t \hat{a}_{t+1} + D_\pi \mathbb{E}_t \hat{a}_{t+2} = \rho_a C_\pi \hat{a}_t + D_\pi \hat{a}_{t+2}$$

Plug the projected solution into the IS curve (A.24) and Phillips curve (A.27) and solve for coefficients using the method of undetermined coefficients,

$$A_y = \frac{\theta s_g}{1 + \Psi \alpha_\pi \Delta(\varphi - 1)} > 0$$

$$A_\pi = \frac{\Delta(\varphi - 1) \theta s_g}{1 + \Psi \alpha_\pi \Delta(\varphi - 1)} > 0$$

$$B_y = \frac{\partial \hat{y}_t}{\partial \hat{\beta}_{t+1}} = -\frac{\Psi}{[1 + \Psi \alpha_\pi \Delta(\varphi - 1)]} < 0$$

$$B_\pi = \frac{\partial \hat{\pi}_t}{\partial \hat{\beta}_{t+1}} = -\frac{\Psi \Delta(\varphi - 1)}{[1 + \Psi \alpha_\pi \Delta(\varphi - 1)]} < 0$$

$\mathbb{I}(\mathbf{H}(\pi_{it}))$ indicates high inflation US-states-years,

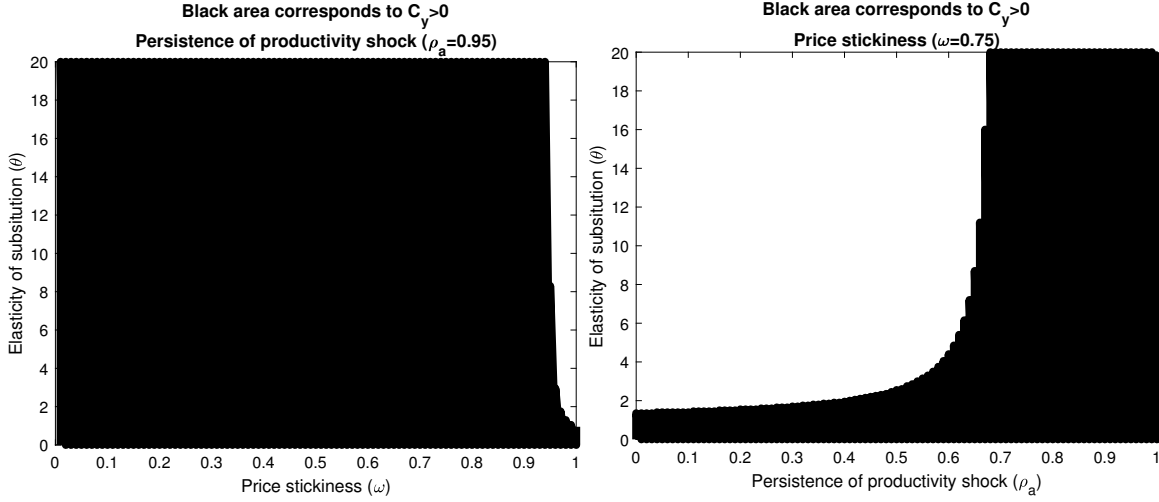
$$D_\pi = 0$$

$$D_y = 0$$

$$C_\pi = \frac{\partial \hat{\pi}_t}{\partial \hat{a}_t} = \frac{-\Delta}{(1 - \beta \rho_a)} \left[\frac{(\varphi - 1)(\theta - 1)(1 - \rho_a)(1 - \beta \rho_a) + \varphi(1 - \rho_a)(1 - \beta \rho_a)}{(1 - \rho_a)(1 - \beta \rho_a) + \Psi(\alpha_\pi - \rho_a)\Delta(\varphi - 1)} \right] < 0$$

$$C_y = \frac{\partial \hat{y}_t}{\partial \hat{a}_t} = \frac{-(\theta - 1)(1 - \rho_a)(1 - \beta \rho_a) + \frac{\theta \varphi(1 - s_g) - (\theta - 1)}{\sigma \varphi} (\alpha_\pi - \rho_a) \frac{(1 - \omega)(1 - \omega \beta)}{\omega} \varphi}{(1 - \rho_a)(1 - \beta \rho_a) + \frac{\theta \varphi(1 - s_g) - (\theta - 1)}{\sigma \varphi} (\alpha_\pi - \rho_a) \frac{(1 - \omega)(1 - \omega \beta)}{\omega} (\varphi - 1)}.$$

Figure A.1: Parameter space corresponding to positive C_y



Notes: The left panel shows the parameter space (θ, ω) that corresponds to positive C_y given the persistence of productivity shock is 0.95. The right panel shows the combination of (θ, ρ_a) that ensures positive C_y .

The sign of all coefficients except C_y is determinant under common parameter values.¹ However, depending on the parameter values, the sign of C_y changes. For example, for a high enough elasticity of substitution (θ) and price-stickiness parameter (ω) or a low enough persistence of productivity shock (ρ_a), C_y can be negative. To determine the sign of C_y , we fix the typical parameter values – the discount factor (β) is 0.99, the Frisch elasticity ($\frac{1}{\varphi-1}$) is 0.5, and coefficient on inflation in the monetary policy rule (α_π) is 1.5. The steady-state government spending to output ratio s_g is calibrated to 0.2. The left panel of Figure A.1 shows the parameter space of θ and ω that corresponds to positive C_y , under the persistence productivity shock (ρ_a) being 0.95. C_y is positive for plausible parameter space. In New Keynesian literature, it is common to set ω as 0.75. The price rigidity of posted prices varies from 0.45 to 0.73 from microdata literature (see Nakamura and Steinsson (2013)). The right panel of Figure A.1 shows the combination of θ and ρ_a that ensures positive C_y , when the price stickiness parameter, ω , is 0.75. For a high enough persistent productivity, we find that C_y is positive. To summarize, C_y is positive under the plausible parameter space. $\square$

Proposition A.2. In a model without DNWR, the government spending multiplier takes the same value M_y in expansion and recession states, i.e. is acyclical.

¹The elasticity of substitution parameter θ is greater than 1, the discount factor β is less than 1 and greater than zero. The government spending share in output, s_g is less than one. The intertemporal elasticity of substitution σ is assumed to be greater than one, while the frequency of price adjustment is ω is less than one. The coefficient on inflation in the monetary policy rule is assumed to be higher than one.

Proof. From the proof of Proposition A.1, the government spending multiplier is

$$M_y \equiv \frac{dy}{dg} = \frac{\partial \hat{y}_t}{\partial \hat{g}_t} \frac{\bar{y}}{\bar{g}} = \frac{A_y}{s_g} = \frac{\omega \theta}{\omega + \Psi \alpha_\pi (1 - \omega)(1 - \omega \beta)(\varphi - 1)} \geq 0$$

regardless of the shock processes and the state of the economy. $\square$

Proposition A.3. When DNWR binds in period t under the expectation of achieving full employment in period $(t + 1)$, the spending multiplier is M_{DNWR} , which is bigger than M_y – the multiplier when DNWR does not bind.

Proof. Guess the solution that satisfies both IS curve (Equation (A.24)) and the modified Phillips curve (Equation (A.29)). Note that the binding DNWR constraint leaves IS curve unchanged while PC changes. Let's first consider the demand-driven business cycle – ($\mathbb{E}_t \hat{\beta}_{t+1} = \hat{\beta}_{t+1}$, and $\mathbb{E}_t \hat{\beta}_{t+2} = 0$). Then, the projected solution becomes

$$\hat{y}_t = F_y \hat{w}_{t-1} + H_y \hat{g}_t + I_y \mathbb{E}_t \hat{\beta}_{t+1} \quad (\text{A.31})$$

$$\hat{\pi}_t = F_\pi \hat{w}_{t-1} + H_\pi \hat{g}_t + I_\pi \mathbb{E}_t \hat{\beta}_{t+1}. \quad (\text{A.32})$$

Under the assumption that DNWR does not bind in period $(t+1)$, the expected output and inflation $\mathbb{E}_t \hat{y}_{t+1}$ and $\mathbb{E}_t \hat{\pi}_{t+1}$ become zero. Plug in suggested solutions (A.31) and (A.32) into IS curve (A.24) and the modified Phillips curves (Equation (A.29)) and find the coefficients using the method of undetermined coefficients,

$$F_y \hat{w}_{t-1} + H_y \hat{g}_t + I_y \hat{\beta}_{t+1} = \theta s_g \hat{g}_t - \Psi \alpha_\pi (F_\pi \hat{w}_{t-1} + H_\pi \hat{g}_t + I_\pi \hat{\beta}_{t+1}) - \Psi \hat{\beta}_{t+1}$$

$$(1 + \Delta)(F_\pi \hat{w}_{t-1} + H_\pi \hat{g}_t + I_\pi \hat{\beta}_{t+1}) = \Delta \hat{w}_{t-1}$$

The multiplier in the demand-driven business cycle is

$$M_{DNWR}^D = \frac{dy}{dg} = \frac{\partial \hat{y}_t}{\partial \hat{g}_t} \frac{y}{g} = H_y \frac{1}{s_g} = \theta$$

, which is bigger than $M_y = \frac{\omega \theta}{\omega + \Psi \alpha_\pi (1 - \omega)(1 - \omega \beta)(\varphi - 1)}$.

Now, let's consider the supply-driven business cycles following ($\hat{a}_t = \hat{a}_t$, $\mathbb{E}_t \hat{a}_{t+1} = \rho_a \hat{a}_t$, and $\mathbb{E}_t \hat{a}_{t+2} = \hat{a}_{t+2}$). Conjecture solution as,

$$\hat{y}_t = O_y \hat{w}_{t-1} + S_y \hat{g}_t + U_y \hat{a}_t + V_y \rho_a \hat{a}_t \quad (\text{A.33})$$

$$\hat{\pi}_t = O_\pi \hat{w}_{t-1} + S_\pi \hat{g}_t + U_\pi \hat{a}_t + V_\pi \rho_a \hat{a}_t. \quad (\text{A.34})$$

Under the assumption that DNWR does not bind in period $(t+1)$, the expected output and inflation are given by the full employment solution shown in the proof of Proposition A.1, as below.

$$\mathbb{E}_t \hat{y}_{t+1} = C_y \rho_a \hat{a}_t + D_y \hat{a}_{t+2} \quad (\text{A.35})$$

$$\mathbb{E}_t \hat{\pi}_{t+1} = C_\pi \rho_a \hat{a}_t + D_\pi \hat{a}_{t+2} \quad (\text{A.36})$$

Combining the suggested solution ((A.33) and (A.34)) with the expected output and inflation ((A.35) and (A.36)) into the IS curve (A.24) and the modified Phillips curves (Equation (A.29)) brings

$$\begin{aligned} O_y \hat{w}_{t-1} + S_y \hat{g}_t + U_y \hat{a}_t + V_y \rho_a \hat{a}_t &= C_y \rho_a \hat{a}_t + D_y \hat{a}_{t+2} - (\theta - 1)(\hat{a}_t - \rho_a \hat{a}_t) \\ &\quad + \theta s_g \hat{g}_t - \Psi \alpha_\pi (O_\pi \hat{w}_{t-1} + S_\pi \hat{g}_t + U_\pi \hat{a}_t + V_\pi \rho_a \hat{a}_t) + \Psi (C_\pi \rho_a \hat{a}_t + D_\pi \hat{a}_{t+2}) \end{aligned}$$

$$(1 + \Delta)[O_\pi \hat{w}_{t-1} + S_\pi \hat{g}_t + U_\pi \hat{a}_t + V_\pi \rho_a \hat{a}_t] = \Delta[\hat{w}_{t-1} - \hat{a}_t] + \beta(C_\pi \rho_a \hat{a}_t + D_\pi \hat{a}_{t+2})$$

Using the undetermined coefficients method, we find $S_\pi = 0$ and $S_y = \theta s_g$. The output multiplier in the supply-driven business cycle is

$$M_{DNWR}^S = \frac{\partial \hat{y}}{\partial \hat{g}} \frac{y}{g} = S_y \frac{1}{s_g} = \theta.$$

Thus, we have shown that the multiplier is θ when DNWR binds (M_{DNWR}), regardless of the source of fluctuation. $\square$

Lemma A.1. Assume the economy is at the steady-state in period $t - 1$, $\hat{w}_{t-1} = 0$. In the presence of the DNWR constraint ($\gamma = 1$), a positive discount factor shock or a negative productivity shock triggers the DNWR constraint to bind and induces unemployment in period t .

Proof. Log-linearized DNWR constraint (Equation (A.4)) can be expressed as follows.

$$\hat{w}_t \geq \gamma(\hat{w}_{t-1} - \hat{\pi}_t). \quad (\text{A.37})$$

To show the DNWR constraint binds in period t under the assumption that $\hat{w}_{t-1} = 0$ and $\gamma = 1$, we have to show

$$\hat{w}_t + \hat{\pi}_t < 0. \quad (\text{A.38})$$

Let's conjecture DNWR does not bind and $\hat{n}_t = \hat{n}_t^s$. Now check whether the conjecture

holds, that is, Equation (A.37) is true. First, we obtain $\hat{w}_t$ by combining two log-linearized Equation (A.2) and (A.12):

$$\hat{w}_t = (\varphi - 1)(\hat{y}_t - \hat{a}_t).$$

From the proof of Proposition A.1, we know that we can write $\hat{y}_t$ and $\hat{\pi}_t$ as follows.

$$\hat{y} = A_y \hat{g}_t + B_y \mathbb{E}_t \hat{\beta}_{t+1} + C_y \hat{a}_t + D_y \mathbb{E}_t \hat{a}_{t+1} = B_y \mathbb{E}_t \hat{\beta}_{t+1} + C_y \hat{a}_t + \rho_a D_y \hat{a}_t$$

$$\hat{\pi} = A_\pi \hat{g}_t + B_\pi \mathbb{E}_t \hat{\beta}_{t+1} + C_\pi \hat{a}_t + D_\pi \mathbb{E}_t \hat{a}_{t+1} = B_\pi \mathbb{E}_t \hat{\beta}_{t+1} + C_\pi \hat{a}_t + \rho_a D_\pi \hat{a}_t$$

Plug in $\hat{y}_t$ and $\hat{\pi}_t$ into the left-hand-side of inequality constraint (A.38)

$$\hat{w}_t + \hat{\pi}_t = (\varphi - 1)(B_y \mathbb{E}_t \hat{\beta}_{t+1} + (C_y + \rho_a D_y - 1)\hat{a}_t) + B_\pi \mathbb{E}_t \hat{\beta}_{t+1} + (C_\pi + \rho_a D_\pi)\hat{a}_t$$

In a demand-driven recession, where $\mathbb{E}_t \hat{\beta}_{t+1} = \beta_H$ and $\hat{a}_t = 0$,

$$\hat{w}_t + \hat{\pi}_t = ((\varphi - 1)B_y + B_\pi)\beta_H.$$

From the proof of Proposition A.1, we know coefficients B_y and B_π are negative. Thus, for any positive discount factor shock, we know that

$$\hat{w}_t + \hat{\pi}_t < 0,$$

which contradicts the conjecture. Thus, we conclude that DNWR binds in response to a positive discount factor shock.

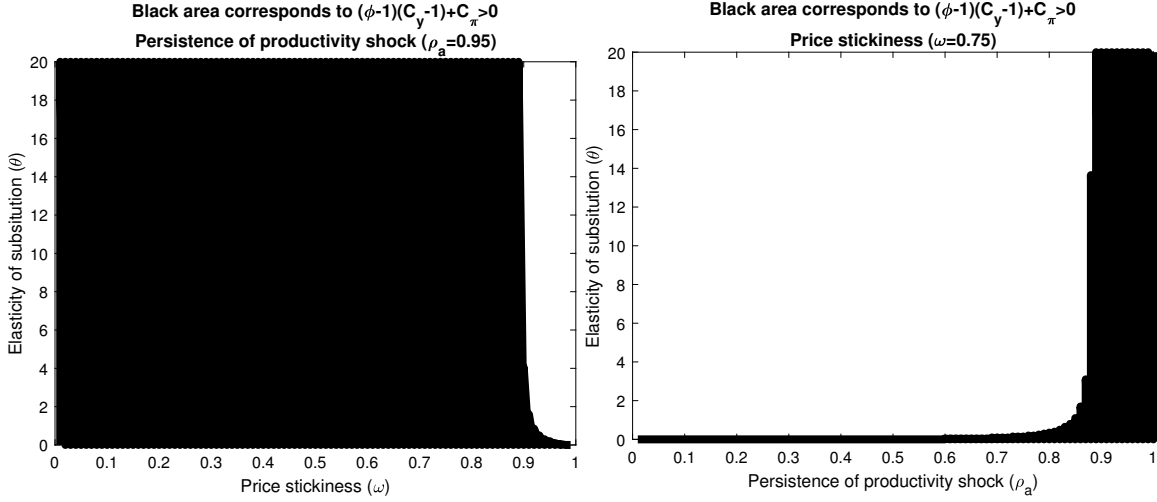
In a supply-driven recession, where $\hat{a}_t = a_L$ and $\mathbb{E}_t \hat{\beta}_{t+1} = 0$,

$$\hat{w}_t + \hat{\pi}_t = (\varphi - 1)((C_y + \rho_a D_y - 1)a_L) + (C_\pi + \rho_a D_\pi)a_L.$$

As $D_y = D_\pi = 0$, the conjecture that DNWR does not bind is not true if

$$\hat{w}_t + \hat{\pi}_t = [(\varphi - 1)(C_y - 1) + C_\pi]a_L < 0.$$

Figure A.2: Parameter space corresponding to positive $(\varphi - 1)(C_y - 1) + C_\pi$



Notes: The left panel shows the parameter space (θ, ω) that gives positive $(\varphi - 1)(C_y - 1) + C_\pi$ given the persistence of productivity shock is 0.95. The right panel shows the combination of (θ, ρ_a) that ensures positive $(\varphi - 1)(C_y - 1) + C_\pi$.

Based on the baseline parameter values², the black area in the left panel of Figure A.2 shows the combination of the elasticity of substitution (θ) and the price stickiness (ω) that satisfies

$$[(\varphi - 1)(C_y - 1) + C_\pi] > 0. \quad (\text{A.39})$$

, where the persistence of the productivity shock ρ_a is 0.95. The right panel of Figure A.2 shows the combination of θ and ρ_a that satisfies Equation (A.39), when the price stickiness parameter, ω , is 0.75. Under the assumption of highly persistent productivity shock, we conclude that DNWR condition binds. $\square$

Lemma A.2. Assume the economy is at steady-state in period $t-1$, $\hat{w}_{t-1} = 0$. In a demand-driven recession, if government spending is less than $\frac{\Psi}{\theta_{sg}}\beta_H \equiv c_d(\beta_H)$, the DNWR constraint binds, and unemployment is greater than zero. Otherwise, DNWR is no longer a binding constraint, and unemployment is zero. In a supply-driven recession, if government spending is less than $c_s(a_L)$, the DNWR constraint binds, and unemployment is greater than zero. Otherwise, DNWR is no longer a binding constraint, and unemployment is zero.

Proof. Find the upper bound of nonzero $\hat{g}_t$ that still violates DNWR condition, that is, $\hat{w}_t < \gamma(\hat{w}_{t-1} - \hat{\pi}_t)$, or $\hat{w}_t + \hat{\pi}_t < 0$. With the nonzero government spending $\hat{g}_t$, we can guess

²The discount factor (β) is 0.99, the Frisch elasticity $(\frac{1}{\varphi-1})$ is 0.5, and the Taylor coefficient on inflation (α_π) is 1.5.

the solution as

$$\hat{y} = A_y \hat{g}_t + B_y \mathbb{E}_t \hat{\beta}_{t+1} + C_y \hat{a}_t + D_y \mathbb{E}_t \hat{a}_{t+1} = A_y \hat{g}_t + B_y \mathbb{E}_t \hat{\beta}_{t+1} + C_y \hat{a}_t + \rho_a D_y \hat{a}_t$$

$$\hat{\pi} = A_\pi \hat{g}_t + B_\pi \mathbb{E}_t \hat{\beta}_{t+1} + C_\pi \hat{a}_t + D_\pi \mathbb{E}_t \hat{a}_{t+1} = A_\pi \hat{g}_t + B_\pi \mathbb{E}_t \hat{\beta}_{t+1} + C_\pi \hat{a}_t + \rho_a D_\pi \hat{a}_t.$$

Then we can rewrite the left-hand-side of the DNWR constraint (A.38)

$$\hat{w}_t + \hat{\pi}_t = (\varphi - 1)(A_y \hat{g}_t + B_y \mathbb{E}_t \hat{\beta}_{t+1} + (C_y + \rho_a D_y - 1)\hat{a}_t) + A_\pi \hat{g}_t + B_\pi \mathbb{E}_t \hat{\beta}_{t+1} + (C_\pi + \rho_a D_\pi)\hat{a}_t$$

In a demand-driven recession, where $\mathbb{E}_t \hat{\beta}_{t+1} = \beta_H$ and $\hat{a}_t = 0$,

$$\hat{w}_t + \hat{\pi}_t = ((\varphi - 1)A_y + A_\pi)\hat{g}_t + ((\varphi - 1)B_y + B_\pi)\beta_H$$

Using the coefficients that we find from the proof of Proposition A.1, we can rewrite the above equation as

$$\hat{w}_t + \hat{\pi}_t = \left(\frac{(\varphi - 1)\theta s_g(1 + \Delta)}{1 + \Psi\alpha_\pi\Delta(\varphi - 1)} \right) \hat{g}_t + \left(-\frac{\Psi(1 + \Delta)(\varphi - 1)}{[1 + \Psi\alpha_\pi\Delta(\varphi - 1)]} \right) \beta_H$$

DNWR binds with non-zero government spending if $\hat{w}_t + \hat{\pi}_t < 0$, that is,

$$\frac{(\varphi - 1)\theta s_g(1 + \Delta)}{1 + \Psi\alpha_\pi\Delta(\varphi - 1)} \hat{g}_t < \frac{\Psi(1 + \Delta)(\varphi - 1)}{1 + \Psi\alpha_\pi\Delta(\varphi - 1)} \beta_H$$

$$\hat{g}_t < \frac{\Psi}{\theta s_g} \beta_H \equiv c_d(\beta_H)$$

In a supply-driven recession, where $\hat{a}_t = a_L$ and $\mathbb{E}_t \hat{\beta}_{t+1} = 0$, the left-hand-side of the inequality constraint (A.38) is

$$\hat{w}_t + \hat{\pi}_t = [(\varphi - 1)A_y + A_\pi]\hat{g}_t + [(\varphi - 1)(C_y - 1) + C_\pi]a_L.$$

DNWR binds with non-zero government spending if $\hat{w}_t + \hat{\pi}_t < 0$, or, equivalently,

$$\hat{g}_t < \frac{[(\varphi - 1)(C_y - 1) + C_\pi]}{[(\varphi - 1)A_y + A_\pi]} (-a_L) \equiv c_s(a_L) \quad (\text{A.40})$$

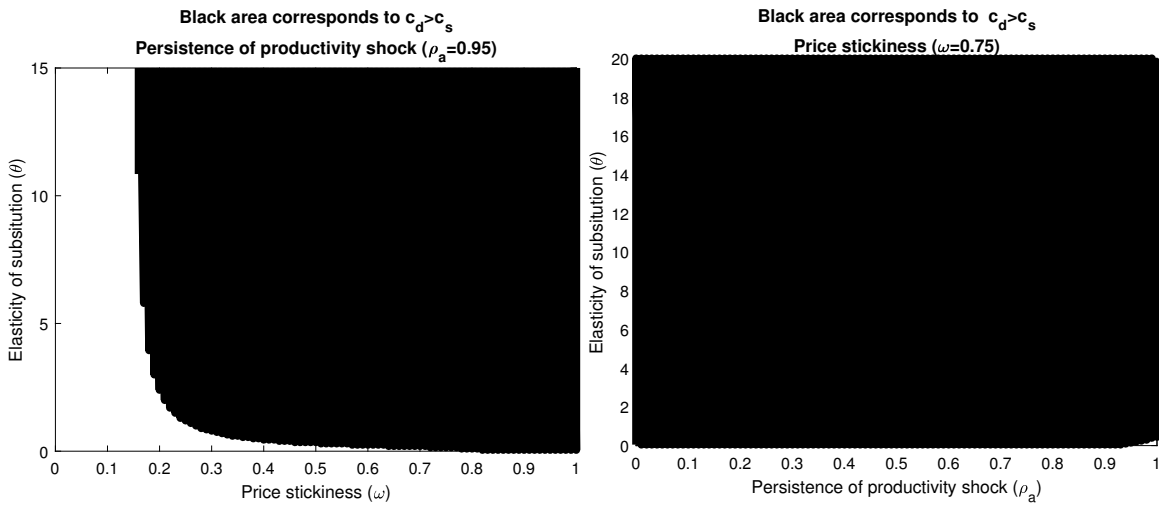
Given the negative productivity shock, the right hand side of Equation (A.40) is positive. Note that we show both A_y and A_π are positive in the proof of Proposition A.1 and $[(\varphi - 1)(C_y - 1) + C_\pi]$ is positive from the proof of Lemma A.1. $\square$

Lemma A.3. Under the assumption that $|\beta_H| = |a_L|$, it can be shown that $0 < c_s(a_L) < c_d(\beta_H)$. In other words, the government spending required to ensure DNWR is no longer binding is smaller in a supply driven recession than a demand driven recession.

Proof. For given $|\beta_H| = |a_L|$, we want to show that

$$\frac{[(\varphi - 1)(C_y - 1) + C_\pi]}{[(\varphi - 1)A_y + A_\pi]} < \frac{\Psi}{\theta s_g} \quad (\text{A.41})$$

Figure A.3: Parameter space corresponding to $c_s(a_L) < c_d(\beta_H)$



Notes: The left panel shows the parameter space (θ, ω) that satisfies $c_s(a_L) < c_d(\beta_H)$ given the persistence of productivity shock is 0.95. The right panel shows the combination of (θ, ρ_a) that ensures $c_s(a_L) < c_d(\beta_H)$.

Based on the baseline parameter values, the black area in the left panel of Figure A.3 shows the combination of the elasticity of substitution (θ) and the price stickiness (ω) that satisfies Equation (A.41), where the persistence of the productivity shock ρ_a is 0.95. The right panel of Figure A.3 shows the combination of θ and ρ_a that satisfies Equation (A.41), when the price stickiness parameter, ω , is 0.75. We find the Equation (A.41) holds for most cases. $\square$

Proposition 1. Under the assumption that $|\beta_H| = |a_L|$, i.e. equal sized business cycle fluctuations,

- the spending multiplier in a demand-driven recession $\geq$
- the spending multiplier in a supply-driven recession $\geq$
- the spending multiplier in an expansion,

for a given size of government spending shock.

Proof. In the absence of DNWR, the multipliers are the same regardless of the state of the economy or the source of fluctuation. In the presence of DNWR ($\gamma = 1$), if government spending (g) satisfies $g < c_s(a_L)$, the DNWR constraint still binds for both recessions (Lemma A.2), thus, the spending multiplier in a demand-driven recession (M_{DNWR}^D) is the same as the spending multiplier in a supply-driven recession (M_{DNWR}^S), which is greater than the spending multiplier in an expansion (M_y). If $c_s(a_L) < g < c_d(\beta_H)$, DNWR condition binds in a demand-driven recession but not in a supply-driven recession. In this case, the spending multiplier in a demand-driven recession is M_{DNWR}^D , which is higher than the spending multiplier in a supply-driven recession when DNWR is not a binding constraint, equal to the spending multiplier in an expansion, M_y . If $c_s(a_L) < c_d(\beta_H) < g$, government spending is large enough to raise nominal wages and achieve full employment, the spending multiplier would be M_y regardless of the source of fluctuation and the state of the business cycle. $\square$

A.2 Appendix: Quantitative Model

A.2.1 Business Cycle fluctuations under supply and demand shocks in the quantitative model

We begin by considering the impulse responses to both contractionary and expansionary supply and demand shocks. The size of the shock is normalized to match the average output gap during the Great Recession. According to the Congressional Budget Office estimates,³ the average output gap from 2008 to 2010 was 4%. We consider productivity and discount factor shocks to match this impact on output in a recession. This results in considering 1.7% deviations from the steady-state value of the discount factor and 2.9% deviations from the steady-state value of productivity. Both shock processes follow AR(1) process, following Equation (4) and Equation (5).⁴

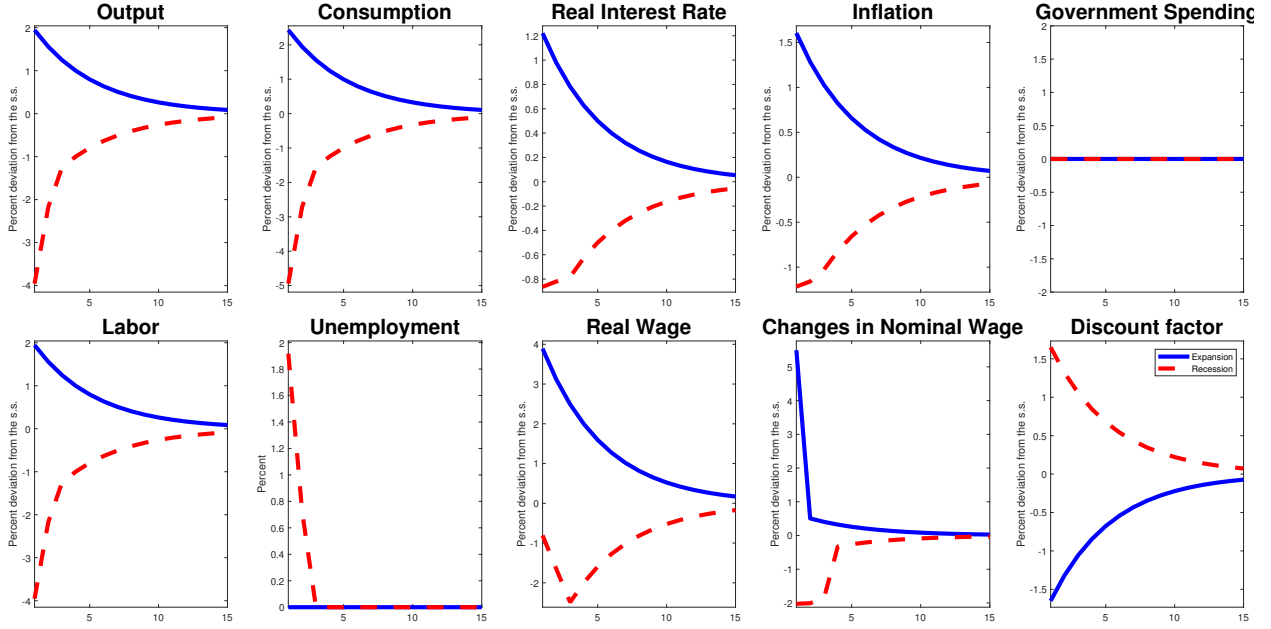
Figure A.4 displays impulse response in a demand-driven expansion and recession, without government spending. In response to a negative discount factor shock (shown with solid blue lines), consumers spend more in the current period leading to a demand-driven expansion. An increase in demand raises inflation and equilibrium labor. As there are no frictions in adjusting nominal wages upward, the labor market always clear, and the unemployment rate is zero.

In response to a positive discount factor shock (shown with dashed red lines in Figure A.4), consumers postpone current consumption, which causes a recession. As labor

³Source: <https://fred.stlouisfed.org/graph/?g=f1cZ>.

⁴We determine the size of the shock based on the average size of the output gap during the Great Recession, however, the slow recovery during the Great Recession was not matched in the following exercises.

Figure A.4: Demand-driven business cycle

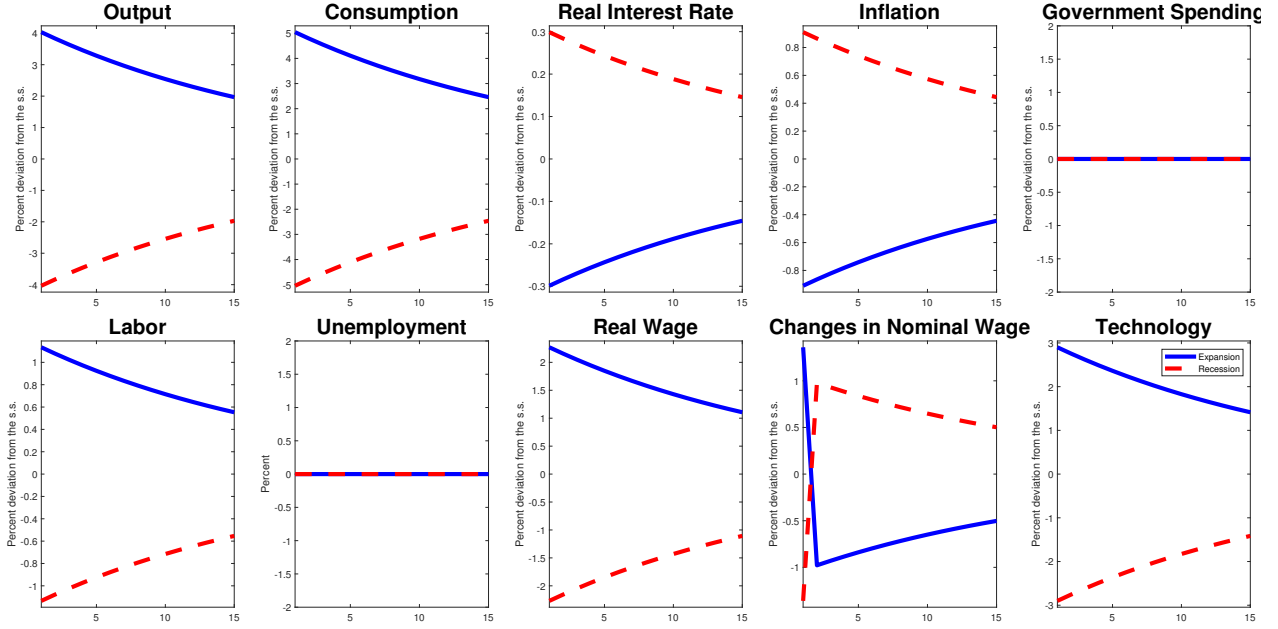


Notes: This graph shows impulse responses to a positive and a negative discount factor shock. The solid blue lines correspond to a negative discount factor shock (a demand-driven expansion), and the dashed red line represents impulse responses to a positive discount factor shock (a demand-driven recession). $\pm 1.7\%$ deviations of the discount factor shocks are imposed. All graph is drawn in terms of the percent deviations from its steady-state except the unemployment rate. The y-axis of the unemployment rate is percent.

demand decreases, there is downward pressure on wages. Although real wage goes up more than 4% in an expansion, the downward adjustment of real wage is about 1% at the beginning of the recession due to deflation and the binding of the DNWR constraint. The DNWR constraint allows at most 2% downward adjustment of real wage. At the same time, there is deflation that drives the real wages upward. The comovement of inflation and output, shown in Proposition A.1, exacerbates the labor market outcome and raises unemployment. Overall, the binding DNWR constraint generates an asymmetric business cycle.

Figure A.5 shows a supply-driven business cycle. As shown in Proposition A.1, inflation and output move in the opposite directions in a supply-driven recession. In a recession (dashed red lines), the marginal product of labor goes down, and firms hire less labor. Accordingly, nominal wage goes down about 1.5%. As we allow the downward adjustment of nominal wage up to 2%, the DNWR constraint does not bind. Consequently, the labor market clears, and the unemployment rate is zero. Unlike the demand-driven recession, the downward adjustment of real wage is greater than that of nominal wage in the supply-driven recession due to inflation. This is also highlighted in the analytical

Figure A.5: Supply-driven business cycle



Notes: This graph displays impulse responses to a positive and a negative productivity shock. The solid blue lines correspond to a positive productivity shock (a supply-driven expansion). The dashed red lines represent impulse responses to a negative productivity shock (a supply-driven recession). $\pm 2.9\%$ deviations of the technology shocks are imposed. All graphs are drawn in terms of the percent deviations from its steady-state except for the unemployment rate. The y-axis of the unemployment rate is percent.

section. The supply-driven business cycle is fully symmetric as DNWR does not bind.⁵

A.2.2 Robustness results

Our baseline model considers GHH (Greenwood, Hercowitz, and Huffman (1988)) preferences which do not allow a wealth effect on labor supply. We relax this assumption and allow for wealth effects on labor supply by introducing KPR (King, Plosser, and Rebelo (1988)) preferences commonly used in the literature. In particular, the preferences take the following form,

$$U(c_t, n_t) = \frac{[c_t(1 - \chi n_t^\varphi)]^{1-\sigma}}{1 - \sigma},$$

where we calibrate φ to ensure the same degree of Frisch elasticity of labor supply as in our baseline model.

As Table 5 shows, the multipliers under these preferences are smaller across the board

⁵When we consider King, Plosser, and Rebelo (1988) preferences in Section A.2.2, we find asymmetric business cycles in response to a supply shock as well. Once we allow for a wealth effect on labor supply in response to a technology shock, nominal wages fall more than in our baseline case and are bound below by DNWR in a supply-driven recession as well.

Table A.1: Cumulative output and consumption multipliers under the KPR preference

		Demand-driven business cycle			Supply-driven business cycle			
		Impact	4 quarters	20 quarters				
		KPR preference						
Output	Expansion	0.485	0.485	0.485	Expansion	0.485	0.485	0.485
Multiplier	Recession	0.668	0.621	0.564	Recession	0.528	0.500	0.494
Consumption	Expansion	-0.515	-0.515	-0.515	Expansion	-0.515	-0.515	-0.515
Multiplier	Recession	-0.332	-0.379	-0.436	Recession	-0.472	-0.500	-0.506

Notes. This table reports the cumulative output and consumption multipliers in an expansion and a recession depending on the source of fluctuation. The cumulative output and consumption multipliers are calculated as $\sum_{i=0}^{k-1} \frac{\Delta y_{t+i}}{(1+r_{t+i})} / \sum_{i=0}^{k-1} \frac{\Delta g_{t+i}}{(1+r_{t+i})}$ and $\sum_{i=0}^{k-1} \frac{\Delta c_{t+i}}{(1+r_{t+i})} / \sum_{i=0}^{k-1} \frac{\Delta g_{t+i}}{(1+r_{t+i})}$, respectively, where Δ denotes the level differences of each variable with and without government spending and r_t is the real interest rate.

relative to GHH preferences.⁶ An increase in government spending under KPR preferences leads to negative wealth effects on the labor supply, as agents internalize higher taxes now or in the future. Earlier studies, such as [Monacelli and Perotti \(2008\)](#), have shown that the degree of complementarity between hours and consumption is inversely related to this wealth effect on labor. Thus, this complementarity is largest with GHH preferences, and declines as we move towards KPR preferences, and leads to a larger response of output under GHH preferences. Under KPR preferences, the multiplier in a demand-driven recession is larger than in an expansion (0.67 in a recession and 0.49 in an expansion under KPR preferences), but the difference is much smaller in magnitude relative to under GHH preferences, (1.74 in a recession and 0.54 in an expansion under GHH preferences). This is because the labor supply curve shifts to the right, and overall weakens the effects of increased spending in reducing unemployment. Under these preferences, DNWR binds in a supply-driven recession as well, leading to a larger output multiplier in a recession relative to an expansion.⁷ The multiplier in a supply-driven recession is thus smaller than the multiplier in a demand-driven recession (0.53 versus 0.67, respectively), although this difference across states is small. The intuition follows from Proposition 1 shown in Section 3. We also show robustness results respect to trend inflation ([Appendix A.2.3](#)), and alternative degree of price and wage rigidity ([Appendix A.2.4](#)).

⁶Under these preferences, we need to adjust the size of both the discount factor and productivity shock in order to generate the same size recession state.

⁷DNWR is more likely to bind in this case in response to a technology shock, since wages have a relatively larger response and labor has a smaller response with KPR preferences as the wealth effects from a technology shock shift the labor supply curve, an effect missing with GHH preferences.

A.2.3 Robustness to trend inflation

Table A.2: Cumulative output and consumption multipliers with nonzero steady-state inflation

		Demand-driven business cycle			Supply-driven business cycle		
		Impact	4 quarters	20 quarters	Impact	4 quarters	20 quarters
A. 2% steady-state annual inflation with GHH preference							
Output	Expansion	0.345	0.314	0.236	Expansion	0.345	0.314
Multiplier	Recession	1.067	0.653	0.439	Recession	0.345	0.314
Consumption	Expansion	-0.655	-0.686	-0.764	Expansion	-0.655	-0.686
Multiplier	Recession	0.067	-0.347	-0.561	Recession	-0.655	-0.686
A. 2% steady-state annual inflation with KPR preference							
Output	Expansion	0.477	0.467	0.446	Expansion	0.477	0.467
Multiplier	Recession	0.646	0.598	0.529	Recession	0.512	0.480
Consumption	Expansion	-0.523	-0.533	-0.554	Expansion	-0.523	-0.533
Multiplier	Recession	-0.354	-0.402	-0.471	Recession	-0.488	-0.520

Notes. This table reports the cumulative output and consumption multipliers in an expansion and a recession with 2% steady-state annual inflation under GHH and KPR preferences.. The cumulative output and consumption multipliers are calculated as $\sum_{i=0}^{i=k-1} \frac{\Delta y_{t+i}}{(1+r_{t+i})} / \sum_{i=0}^{i=k-1} \frac{\Delta g_{t+i}}{(1+r_{t+i})}$ and $\sum_{i=0}^{i=k-1} \frac{\Delta c_{t+i}}{(1+r_{t+i})} / \sum_{i=0}^{i=k-1} \frac{\Delta g_{t+i}}{(1+r_{t+i})}$, respectively, where Δ denotes the level differences of each variable with and without government spending and r_t is the real interest rate.

While demand and supply shocks generate deviations from steady-state inflation in opposite direction, we also consider the importance of the level of steady-state inflation. Table A.2 shows the cumulative output and consumption multipliers when we consider a 2% annual steady-state inflation. The main results hold qualitatively: notably that the output and consumption multipliers are higher in a demand-driven recession compared to an expansion for both GHH and KPR preferences. For KPR preferences, shown in the bottom panel, similar to the zero steady-state inflation case, the multiplier in a supply-driven recession, while larger than in an expansion, is smaller than in a demand-driven recession.

The positive steady-state inflation multipliers are overall smaller than the zero steady-state inflation multipliers. This is because non-zero steady-state inflation in a New Keynesian model leads to a rise in price dispersion and a loss in labor efficiency in response to exogenous shocks. With GHH preferences, the increase in inefficient price dispersion due to an increase in government spending limits the expansionary effects on output significantly. As a result, the government spending multipliers are much smaller with a 2% steady-state annual inflation rate. However, with KPR preference, labor supply also responds to government spending shocks which partially offsets the impact on aggregate

demand due to a change in price dispersion. Consequently, there are smaller differences in the size of the multipliers across zero and non-zero steady-state inflation.

A.2.4 Robustness to alternative degree of nominal rigidity

We also consider an alternative degree of downward nominal wage rigidity and price rigidity. Once we assume a more downwardly flexible wage ($\gamma = 0.96$), it results in a lower multiplier in a demand-driven recession, as shown in Panel A of Table A.3. In contrast, a more rigid wage rigidity assumption ($\gamma = 0.99$) leads to higher multipliers in both recessions than the baseline case, reported in Panel B of Table A.3. The main results still hold that the multiplier in a demand-driven recession is higher than the multipliers in a supply-driven recession and expansion. These results confirm that the extent of binding DNWR is one of the key determinants of the size of multipliers.

Table A.3: Cumulative output and consumption multipliers by the source of fluctuation under alternative degree of wage and price rigidity

		Demand-driven business cycle			Supply-driven business cycle		
		Impact	4 quarters	20 quarters	Impact	4 quarters	20 quarters
A. Less rigid DNWR ($\gamma = 0.96$)							
Output	Expansion	0.535	0.535	0.535	Expansion	0.535	0.535
Multiplier	Recession	1.124	0.733	0.649	Recession	0.535	0.535
Consumption	Expansion	-0.465	-0.465	-0.465	Expansion	-0.465	-0.465
Multiplier	Recession	0.124	-0.267	-0.351	Recession	-0.465	-0.465
B. More rigid DNWR ($\gamma = 0.99$)							
Output	Expansion	0.535	0.535	0.535	Expansion	0.535	0.535
Multiplier	Recession	3.046	2.683	1.849	Recession	1.128	0.734
Consumption	Expansion	-0.465	-0.465	-0.465	Expansion	-0.465	-0.465
Multiplier	Recession	2.046	1.683	0.849	Recession	0.128	-0.350
C. Less rigid prices ($\omega = 0.65$)							
Output	Expansion	0.259	0.259	0.259	Expansion	0.259	0.259
Multiplier	Recession	1.868	1.067	0.727	Recession	0.259	0.259
Consumption	Expansion	-0.741	-0.741	-0.741	Expansion	-0.741	-0.741
Multiplier	Recession	0.868	0.067	-0.273	Recession	-0.741	-0.741
D. More rigid prices ($\omega = 0.85$)							
Output	Expansion	1.269	1.269	1.269	Expansion	1.269	1.269
Multiplier	Recession	2.397	1.968	1.674	Recession	1.269	1.269
Consumption	Expansion	0.269	0.269	0.269	Expansion	0.269	0.269
Multiplier	Recession	1.397	0.968	0.674	Recession	0.269	0.269

Notes. This table reports the cumulative output and consumption multipliers in an expansion and a recession depending on the source of fluctuation. The cumulative output and consumption multipliers are calculated as $\sum_{i=0}^{k-1} \frac{\Delta y_{t+i}}{(1+r_{t+i})} / \sum_{i=0}^{k-1} \frac{\Delta g_{t+i}}{(1+r_{t+i})}$ and $\sum_{i=0}^{k-1} \frac{\Delta c_{t+i}}{(1+r_{t+i})} / \sum_{i=0}^{k-1} \frac{\Delta g_{t+i}}{(1+r_{t+i})}$, respectively, where Δ denotes the level differences of each variable with and without government spending and r_t is the real interest rate.

When considering the dynamics of real wages in a recession, the degree of price rigid-

ity also matters in determining the government spending multiplier. Panel C and D of Table A.3 reports the government spending multipliers with less and more rigid prices than the benchmark case, respectively. Overall, the government spending multipliers are larger in an economy with higher price rigidity, which is seen in a standard New Keynesian model also. With higher price rigidity, an increase in spending raises prices less and labor demand shifts out more due to increased public spending demand, leading to a larger increase in output. This increased price rigidity combined with GHH preferences in Panel C lead to an output multiplier larger than 1, even in an expansion. However, in our specific simulations, price rigidities also matter for the real wage dynamics with DNWR binding. Our results that the spending multipliers are higher in a demand-driven recession are robust for different degrees of price rigidity. In our baseline case, the multiplier in a demand recession is close to 70% larger than in a supply driven recession. With less rigid prices, the multiplier is over 85% larger, since a lower degree of price rigidity further amplifies the difference in real wage response when the DNWR is binding or not. The difference in the multipliers across the two types of recessions shrinks when we have more rigid prices, where the demand recession multiplier is about 45% larger than a supply recession/ expansion multiplier.

A.3 Appendix: Time series empirical evidence

Our data set constitutes of quarterly data for the U.S. spanning 1889Q1-2015Q4. We define inflation as year-over-year growth of the GDP deflator, and use data for GDP, unemployment rate, government spending and GDP deflator from [Ramey and Zubairy \(2018\)](#). Our baseline measure of narrative military news variables also comes from [Ramey and Zubairy \(2018\)](#). When we use taxes as a control variable, we use tax revenues as a share of GDP as a control. This series also comes from [Ramey and Zubairy \(2018\)](#).

When we consider real interest rate as a control variable, we construct the real rate as 3 month T-bill rate minus YoY GDP deflator inflation. Since the interpolated series for the T-bill rate is available only from 1915q1 onward, the regressions with real interest rates as controls are run on a shorter sample than our baseline results.

Table A.4: State-dependent fiscal multipliers for output: military news shocks (Robustness to potential GDP)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	2-year integral output				4-year integral output			
Σg_t	0.67 (0.05)				0.71 (0.03)			
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))$		0.64 (0.08)		0.64 (0.08)		0.68 (0.10)		0.68 (0.10)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))$		0.61 (0.09)				0.69 (0.05)		
$\Sigma g_t \times \mathbb{I}(L(\pi_{t-1}))$			0.66 (0.09)				0.63 (0.07)	
$\Sigma g_t \times \mathbb{I}(H(\pi_{t-1}))$			0.65 (0.05)				0.75 (0.04)	
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.96 (0.18)				0.78 (0.05)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				-0.13 (0.11)				0.28 (0.06)
P-value from the test								
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))$		0.83				0.95		
$\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(\pi_{t-1}))$			0.88				0.14	
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.13				0.47
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.00				0.00
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.00				0.00
Effective first-stage F statistics								
Linear	20.59				11.55			
$\mathbb{I}(L(u_{t-1}))$		10.23		8.82		15.01		14.58
$\mathbb{I}(H(u_{t-1}))$		382.96				122.45		
$\mathbb{I}(L(\pi_{t-1}))$			10.62				5.24	
$\mathbb{I}(H(\pi_{t-1}))$			72.43				46.82	
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				74.53				305.02
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				52.23				75.99
Observations	493	493	493	493	485	485	485	485

Notes: The top panel reports the 2 and 4 year cumulative multiplier along with associated heteroskedasticity and autocorrelation robust standard errors below. $\mathbb{I}(L(u_{t-1}))$ and $\mathbb{I}(H(u_{t-1}))$ indicate the state where lagged unemployment is low and high, respectively. $\mathbb{I}(L(\pi_{t-1}))$ and $\mathbb{I}(H(\pi_{t-1}))$ represent the state where lagged inflation is low and high, respectively. Potential GDP is constructed using the CBO measure of potential GDP with an interpolated measure of potential GDP for early years. The second panel shows p-values testing whether the multipliers are statistically significantly different across states. The last panel reports the [Olea and Pflueger \(2013\)](#) effective first-stage F statistics for military news as an instrument at 2 and 4 year horizons for the relevant subsample. We use the threshold for the 5 percent critical value for testing the null hypothesis that the TSLS bias exceeds 10 percent of the OLS bias. For one instrument, this threshold is always 23.1.

Table A.5: State-dependent fiscal multipliers: military news shocks (Robustness to inflation thresholds: 3%)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	2-year integral output				4-year integral output			
Σg_t	0.66 (0.07)				0.71 (0.04)			
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))$		0.59 (0.09)		0.59 (0.09)		0.67 (0.12)		0.67 (0.12)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))$		0.60 (0.09)				0.68 (0.05)		
$\Sigma g_t \times \mathbb{I}(L(\pi_{t-1}))$			0.83 (0.12)				0.73 (0.08)	
$\Sigma g_t \times \mathbb{I}(H(\pi_{t-1}))$			0.61 (0.05)				0.71 (0.04)	
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				1.07 (0.22)				0.76 (0.06)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.42 (0.13)				0.59 (0.07)
P-value from the test								
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))$		0.95				0.92		
$\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(\pi_{t-1}))$			0.10				0.84	
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.07				0.55
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.25				0.56
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.01				0.06
Effective first-stage F statistics								
Linear	19.38				11.22			
$\mathbb{I}(L(u_{t-1}))$		8.44		8.23		10.85		10.55
$\mathbb{I}(H(u_{t-1}))$		403.28				130.20		
$\mathbb{I}(L(\pi_{t-1}))$			4.90				4.26	
$\mathbb{I}(H(\pi_{t-1}))$			109.31				38.65	
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				350.03				228.48
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				279.98				330.42
Observations	493	493	493	493	485	485	485	485

Notes: The top panel reports the 2 and 4 year cumulative multiplier along with associated heteroskedasticity and autocorrelation robust standard errors below. $\mathbb{I}(L(u_{t-1}))$ and $\mathbb{I}(H(u_{t-1}))$ indicate the state where lagged unemployment is low and high, respectively. $\mathbb{I}(L(\pi_{t-1}))$ and $\mathbb{I}(H(\pi_{t-1}))$ represent the state where lagged inflation is lower and higher than 3%, respectively. The second panel shows p-values testing whether the multipliers are statistically significantly different across states. The last panel reports the [Olea and Pflueger \(2013\)](#) effective first-stage F statistics for military news as an instrument at 2 and 4 year horizons for the relevant subsample. We use the threshold for the 5 percent critical value for testing the null hypothesis that the TSLS bias exceeds 10 percent of the OLS bias. For one instrument, this threshold is always 23.1.

Table A.6: State-dependent fiscal multipliers for output: military news shocks (Robustness to time-varying thresholds)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	2-year integral output				4-year integral output			
Σg_t	0.66 (0.07)				0.71 (0.04)			
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))$		0.66 (0.18)		0.66 (0.18)		0.75 (0.26)		0.75 (0.26)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))$		0.52 (0.08)				0.56 (0.08)		
$\Sigma g_t \times \mathbb{I}(L(\pi_{t-1}))$			0.78 (0.11)				0.69 (0.08)	
$\Sigma g_t \times \mathbb{I}(H(\pi_{t-1}))$			0.58 (0.04)				0.73 (0.04)	
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.78 (0.26)				0.58 (0.15)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.26 (0.07)				0.52 (0.15)
P-value from the test								
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))$		0.45				0.51		
$\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(\pi_{t-1}))$			0.10				0.67	
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.71				0.57
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.01				0.45
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.04				0.80
Effective first-stage F statistics								
Linear	19.38				11.22			
$\mathbb{I}(L(u_{t-1}))$		15.36		14.93		5.06		4.92
$\mathbb{I}(H(u_{t-1}))$		3.80				2.75		
$\mathbb{I}(L(\pi_{t-1}))$			6.17				4.40	
$\mathbb{I}(H(\pi_{t-1}))$			131.60				38.59	
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				22.81				9.82
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				19.25				17.83
Observations	493	493	493	493	485	485	485	485

Notes: The top panel reports the 2 and 4 year cumulative multiplier along with associated heteroskedasticity and autocorrelation robust standard errors below. $\mathbb{I}(L(u_{t-1}))$ and $\mathbb{I}(H(u_{t-1}))$ indicate the state where lagged unemployment is low and high, respectively. $\mathbb{I}(L(\pi_{t-1}))$ and $\mathbb{I}(H(\pi_{t-1}))$ represent the state where lagged inflation is low and high, respectively. In this robustness check, we consider time-varying thresholds for both the unemployment rate and inflation. The time-varying trend is based on the HP filter with $\lambda = 10^6$, over a split sample, 1889–1929 and 1947–2015 and linearly interpolated for the small gap in trend unemployment between 1929 and 1947, in order to capture the evolution of the natural rate. The high inflation regime is one where inflation is above a HP filtered trend based on $\lambda = 1600$. The top panel panel reports the 2 and 4 year cumulative multiplier along with associated standard errors below. The second panel shows p-values testing whether the multipliers are statistically significantly different across states. The last panel reports the [Olea and Pflueger \(2013\)](#) effective first-stage F statistics for military news as an instrument at 2 and 4 year horizons for the relevant subsample. We use the threshold for the 5 percent critical value for testing the null hypothesis that the TSLS bias exceeds 10 percent of the OLS bias. For one instrument, this threshold is always 23.1.

Table A.7: State-dependent fiscal multipliers for output: military news shocks (Robustness to time varying thresholds for inflation)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	2-year integral output				4-year integral output			
Σg_t	0.66 (0.07)				0.71 (0.04)			
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))$		0.59 (0.09)		0.59 (0.09)		0.67 (0.12)		0.67 (0.12)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))$		0.60 (0.09)				0.68 (0.05)		
$\Sigma g_t \times \mathbb{I}(L(\pi_{t-1}))$			0.78 (0.11)				0.69 (0.08)	
$\Sigma g_t \times \mathbb{I}(H(\pi_{t-1}))$			0.58 (0.04)				0.73 (0.04)	
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				1.26 (0.27)				0.82 (0.08)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.23 (0.07)				0.58 (0.06)
P-value from the test								
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))$		0.95				0.92		
$\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(\pi_{t-1}))$			0.10				0.67	
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.03				0.34
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.00				0.56
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.00				0.01
Effective first-stage F statistics								
Linear	19.38				11.22			
$\mathbb{I}(L(u_{t-1}))$		8.44		8.23		10.85		10.56
$\mathbb{I}(H(u_{t-1}))$		403.28				130.20		
$\mathbb{I}(L(\pi_{t-1}))$			6.17				4.40	
$\mathbb{I}(H(\pi_{t-1}))$			131.60				38.59	
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				139.80				90.16
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				619.63				722.32
Observations	493	493	493	493	485	485	485	485

Notes: The top panel reports the 2 and 4 year cumulative multiplier along with associated heteroskedasticity and autocorrelation robust standard errors below. $\mathbb{I}(L(u_{t-1}))$ and $\mathbb{I}(H(u_{t-1}))$ indicate the state where lagged unemployment is low and high, respectively. $\mathbb{I}(L(\pi_{t-1}))$ and $\mathbb{I}(H(\pi_{t-1}))$ represent the state where lagged inflation is low and high, respectively. In this robustness check, we consider time-variant thresholds for only inflation. The high inflation regime is one where inflation is above a HP filtered trend based on $\lambda = 1600$. The second panel shows p-values testing whether the multipliers are statistically significantly different across states. The last panel reports the [Olea and Pflueger \(2013\)](#) effective first-stage F statistics for military news as an instrument at 2 and 4 year horizons for the relevant subsample. We use the threshold for the 5 percent critical value for testing the null hypothesis that the TSLS bias exceeds 10 percent of the OLS bias. For one instrument, this threshold is always 23.1.

Table A.8: State-dependent fiscal multipliers for output: both military news and Blanchard-Perotti (2002) as instruments (Robustness to time varying thresholds for inflation)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	2-year integral output				4-year integral output			
Σg_t	0.42 (0.10)				0.56 (0.08)			
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))$		0.33 (0.11)				0.39 (0.11)		
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))$		0.62 (0.09)				0.68 (0.05)		
$\Sigma g_t \times \mathbb{I}(L(\pi_{t-1}))$			0.47 (0.09)				0.49 (0.10)	
$\Sigma g_t \times \mathbb{I}(H(\pi_{t-1}))$			0.48 (0.07)				0.67 (0.07)	
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.43 (0.15)				0.30 (0.20)
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.42 (0.07)				0.55 (0.12)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.92 (0.23)				0.81 (0.07)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.56 (0.20)				0.80 (0.13)
P-value from the test								
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))$		0.10				0.02		
$\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(\pi_{t-1}))$			0.94				0.09	
$\mathbb{I}(L(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(L(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.93				0.21
$\mathbb{I}(L(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.60				0.06
$\mathbb{I}(L(u_{t-1}))\mathbb{I}(H(\pi_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.05				0.06
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.22				0.96
Effective first-stage F statistics								
Linear	36.70 13.19				14.44 15.46			
$\mathbb{I}(L(u_{t-1}))$		48.92 [17.04]				39.63 [17.05]		
$\mathbb{I}(H(u_{t-1}))$		61.56 [19.27]				69.75 [10.83]		
$\mathbb{I}(L(\pi_{t-1}))$			36.59 [20.81]				18.75 [15.56]	
$\mathbb{I}(H(\pi_{t-1}))$			80.18 [15.24]				24.85 [18.13]	
$\mathbb{I}(L(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				51.67 [17.49]				20.15 [15.63]
$\mathbb{I}(L(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				54.60 [7.57]				23.38 [10.05]
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				48.93 [16.48]				33.64 [17.50]
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				53.41 [20.99]				46.18 [21.11]
Observations	493	493	493	493	485	485	485	485

Notes: The top panel reports the 2 and 4 year cumulative multiplier along with associated heteroskedasticity and autocorrelation robust standard errors below. $\mathbb{I}(L(u_{t-1}))$ and $\mathbb{I}(H(u_{t-1}))$ indicate the state where lagged unemployment is low and high, respectively. $\mathbb{I}(L(\pi_{t-1}))$ and $\mathbb{I}(H(\pi_{t-1}))$ represent the state where lagged inflation is low and high, respectively. In this robustness check, we consider time-variant thresholds for only inflation. The high inflation regime is one where inflation is above a HP filtered trend based on $\lambda = 1600$. The second panel shows p-values testing whether the multipliers are statistically significantly different across states. The last panel reports the [Olea and Pflueger \(2013\)](#) effective first-stage F statistics for

Table A.9: State-dependent fiscal multipliers for output: military news shocks (Controlling for taxes)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	2-year integral output				4-year integral output			
Σg_t	0.66 (0.07)				0.72 (0.05)			
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))$		0.54 (0.11)		0.54 (0.11)		0.60 (0.15)		0.60 (0.15)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))$		0.67 (0.12)				0.69 (0.08)		
$\Sigma g_t \times \mathbb{I}(L(\pi_{t-1}))$			0.79 (0.14)				0.68 (0.10)	
$\Sigma g_t \times \mathbb{I}(H(\pi_{t-1}))$			0.54 (0.11)				0.61 (0.10)	
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.99 (0.20)				0.80 (0.06)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				-0.12 (0.12)				0.24 (0.08)
P-value from the test								
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))$		0.47				0.64		
$\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(\pi_{t-1}))$			0.18				0.56	
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.07				0.27
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.00				0.03
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.00				0.00
Effective first-stage F statistics								
Linear	18.73				11.56			
$\mathbb{I}(L(u_{t-1}))$		8.41		8.12		15.98		15.51
$\mathbb{I}(H(u_{t-1}))$		118.25				72.91		
$\mathbb{I}(L(\pi_{t-1}))$			8.07				4.59	
$\mathbb{I}(H(\pi_{t-1}))$			58.30				100.57	
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				72.26				222.68
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				42.84				75.31
Observations	493	493	493	493	485	485	485	485

Notes: The top panel reports the 2 and 4 year cumulative multiplier along with associated heteroskedasticity and autocorrelation robust standard errors below. $\mathbb{I}(L(u_{t-1}))$ and $\mathbb{I}(H(u_{t-1}))$ indicate the state where lagged unemployment is low and high, respectively. $\mathbb{I}(L(\pi_{t-1}))$ and $\mathbb{I}(H(\pi_{t-1}))$ represent the state where lagged inflation is low and high, respectively. These are defined as our baseline specification, shown in Table 2. We include average tax rates as additional controls on the right hand side.

Table A.10: State-dependent fiscal multipliers for output: military news shocks (1915-2015)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	2-year integral output				4-year integral output			
Σg_t	0.70 (0.08)				0.74 (0.05)			
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))$		0.69 (0.08)		0.69 (0.08)		0.68 (0.14)		0.68 (0.14)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))$		0.55 (0.12)				0.68 (0.07)		
$\Sigma g_t \times \mathbb{I}(L(\pi_{t-1}))$			0.74 (0.15)				0.70 (0.12)	
$\Sigma g_t \times \mathbb{I}(H(\pi_{t-1}))$			0.61 (0.07)				0.69 (0.05)	
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.63 (0.30)				0.73 (0.12)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				-0.62 (0.46)				0.03 (0.30)
P-value from the test								
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))$		0.31				0.98		
$\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(\pi_{t-1}))$			0.46				0.94	
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.84				0.80
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.00				0.04
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.02				0.03
Effective first-stage F statistics								
Linear	12.68				9.94			
$\mathbb{I}(L(u_{t-1}))$		6.40		6.17		7.92		7.54
$\mathbb{I}(H(u_{t-1}))$		242.24				76.60		
$\mathbb{I}(L(\pi_{t-1}))$			5.53				4.10	
$\mathbb{I}(H(\pi_{t-1}))$			37.96				45.29	
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				61.74				109.23
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				31.37				41.40
Observations	393	393	393	393	385	385	385	385

Notes: The top panel reports the 2 and 4 year cumulative multiplier along with associated heteroskedasticity and autocorrelation robust standard errors below. $\mathbb{I}(L(u_{t-1}))$ and $\mathbb{I}(H(u_{t-1}))$ indicate the state where lagged unemployment is low and high, respectively. $\mathbb{I}(L(\pi_{t-1}))$ and $\mathbb{I}(H(\pi_{t-1}))$ represent the state where lagged inflation is low and high, respectively. These are defined as our baseline specification, shown in Table 2. The sample under consideration is shorter than our baseline to make comparison with the specification with real interest rates as additional control easier, and spans 1915-2015.

Table A.11: State-dependent fiscal multipliers for output: military news shocks (1915-2015, Controlling for real interest rates)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	2-year integral output				4-year integral output			
Σg_t	0.70 (0.08)				0.75 (0.07)			
$\Sigma g_t \times \mathbb{I}(L(u_{t-1}))$		0.69 (0.14)		0.69 (0.12)		0.77 (0.14)		0.77 (0.14)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))$		0.51 (0.13)				0.64 (0.09)		
$\Sigma g_t \times \mathbb{I}(L(\pi_{t-1}))$			0.73 (0.16)				0.68 (0.13)	
$\Sigma g_t \times \mathbb{I}(H(\pi_{t-1}))$			0.65 (0.05)				0.74 (0.03)	
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.54 (0.28)				0.67 (0.13)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				-0.28 (0.37)				0.11 (0.22)
P-value from the test								
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))$		0.34				0.36		
$\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(\pi_{t-1}))$			0.66				0.65	
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				0.61				0.59
$\mathbb{I}(L(u_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.01				0.01
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				0.08				0.03
Effective first-stage F statistics								
Linear	14.53				10.23			
$\mathbb{I}(L(u_{t-1}))$		4.29		5.59		7.72		7.46
$\mathbb{I}(H(u_{t-1}))$		283.90				14.13		
$\mathbb{I}(L(\pi_{t-1}))$			4.83				4.20	
$\mathbb{I}(H(\pi_{t-1}))$			67.13				42.64	
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$				57.33				142.65
$\mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$				34.98				41.72
Observations	392	392	392	392	384	384	384	384

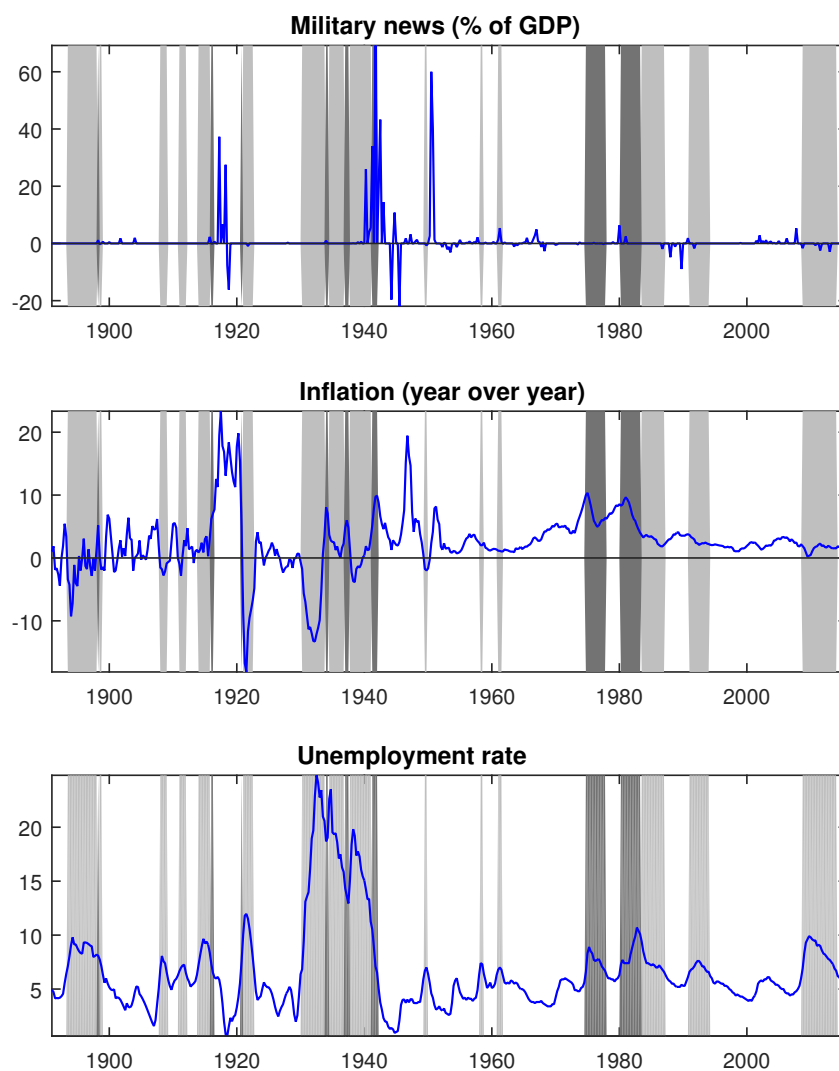
Notes: The top panel reports the 2 and 4 year cumulative multiplier along with associated heteroskedasticity and autocorrelation robust standard errors below. $\mathbb{I}(L(u_{t-1}))$ and $\mathbb{I}(H(u_{t-1}))$ indicate the state where lagged unemployment is low and high, respectively. $\mathbb{I}(L(\pi_{t-1}))$ and $\mathbb{I}(H(\pi_{t-1}))$ represent the state where lagged inflation is low and high, respectively. These are defined as our baseline specification, shown in Table 2. The sample under consideration is shorter than our baseline and spans 1915-2015. We include real interest rates as additional controls on the right hand side.

Table A.12: Response of unemployment, inflation and interest rate

2 year horizon response				
	Unemployment	Inflation	Tbill rate	Real rate
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$	-0.1971 (0.1465)	0.4699 (0.1176)	-0.7946 (3.6384)	-0.5609 (0.1791)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$	0.0784 (0.0833)	0.1599 (0.2370)	8.5678 (19.788)	-0.000 (0.3399)
P-value from test $\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$	0.12	0.23	0.70	0.15
Observation	493	493	389	389
4 year horizon response				
	Unemployment	Inflation	Tbill rate	Real rate
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1}))$	-0.1880 (0.0458)	0.1553 (0.0455)	0.6424 (1.7783)	-0.1716 (0.0458)
$\Sigma g_t \times \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$	0.0133 (0.0530)	0.0963 (0.1479)	-7.5802 (10.947)	0.2179 (0.1634)
P-value from test $\mathbb{I}(H(u_{t-1}))\mathbb{I}(L(\pi_{t-1})) = \mathbb{I}(H(u_{t-1}))\mathbb{I}(H(\pi_{t-1}))$	0.01	0.69	0.46	0.02
Observation	485	485	381	381

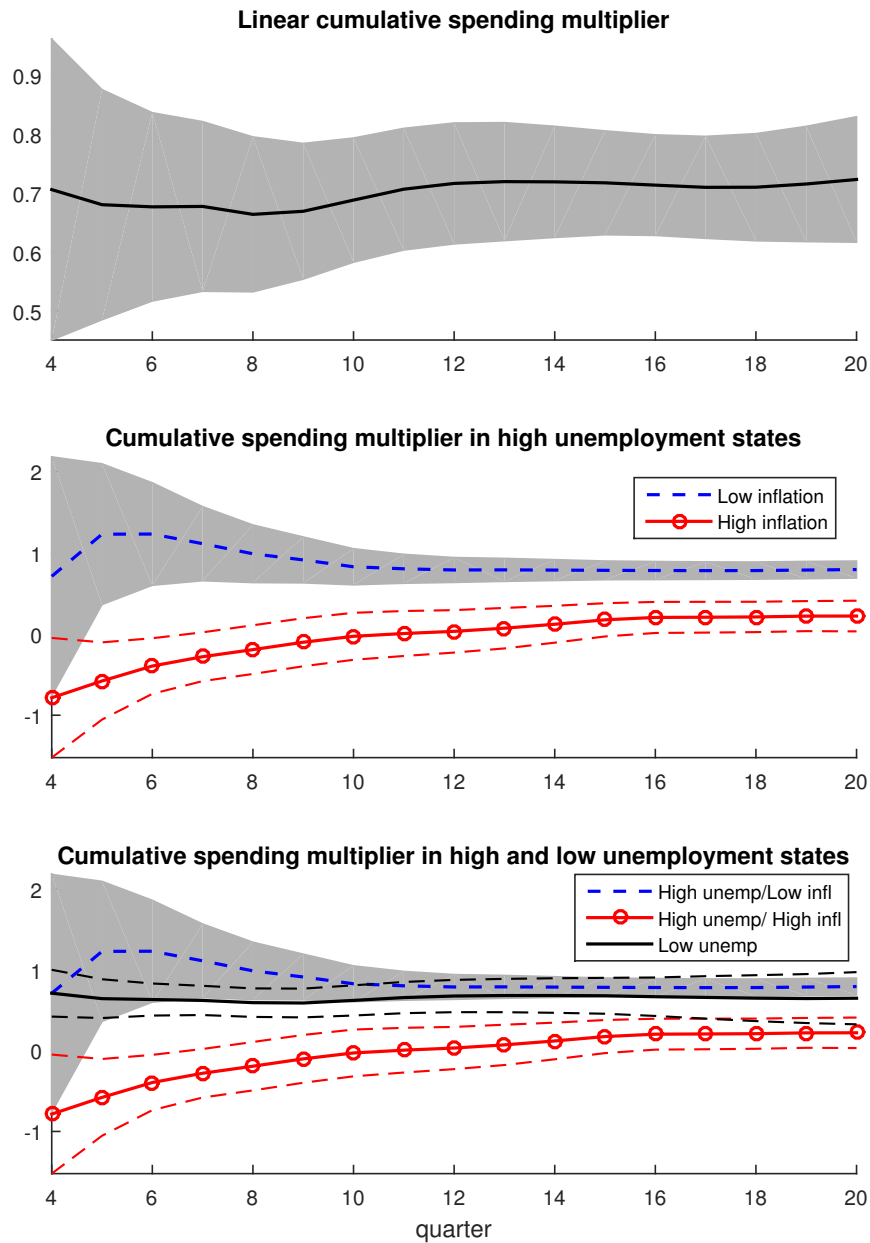
Notes: The top panel reports the 2 and 4 year cumulative responses of unemployment, inflation, Tbill rate, and real interest rate along with associated heteroskedasticity and autocorrelation robust standard errors below. $\mathbb{I}(L(u_{t-1}))$ and $\mathbb{I}(H(u_{t-1}))$ indicate the state where lagged unemployment is low and high, respectively. $\mathbb{I}(L(\pi_{t-1}))$ and $\mathbb{I}(H(\pi_{t-1}))$ represent the state where lagged inflation is low and high, respectively. These are defined as our baseline specification, shown in Table 2.

Figure A.6: Inflation and unemployment states for U.S. historical data



Notes: Military spending news, year over year GDP deflator inflation rate and the unemployment rate. The shaded areas indicate periods when the unemployment rate is above the threshold of 6.5 percent. The light and dark gray areas correspond with periods where inflation is below and above a threshold of 4%, which corresponds to the 75th percentiles of inflation for our full sample.

Figure A.7: State dependent fiscal multipliers: military news shocks



Notes: These figures show the cumulative multiplier for output in response to a military news shock from 4 quarters after the shock hits the economy. The top panel shows the cumulative multiplier in a linear model. The middle panel shows the state-dependent multiplier in high unemployment/ low inflation (blue dashed) and high unemployment/ high inflation (red circles) states. The bottom panel shows the state dependent multipliers in low unemployment (black solid), high unemployment/ low inflation (blue dashed) and high unemployment/ high inflation (red circles) states. 95 percent confidence intervals are shown in all cases.

A.4 Appendix: US state-level empirical evidence

The state-level annual data sample starts in 1969 and ends in 2018. State-level nominal GDP is from the US Bureau of Economic Analysis (BEA). In calculating real GDP, we use the US aggregate Consumer Price Index (CPI) to deflate nominal GDP followed by BEA - calculating state-level GDP by applying national price deflator to state-level nominal GDP. State-level employment is from Current Employment Statistics (CES) by the Bureau of Labor Statistics (BLS) and the state-level population is available from the US Census Bureau. We use state-level inflation data constructed by [Nakamura and Steinsson \(2014\)](#) from 1969 to 2008 and later by [Zidar \(2019\)](#) up to 2014. We further extend the state-level inflation from Regional Price Parity (RPP) from Census until 2018.⁸

For state-level military spending, we use data from prime military contracts awarded by the Department of Defense (DOD). Each individual contractor of DOD reports their contract details using DD Form 350, including the service or product supplies, date awarded, principal place of performance, and information about the DOD agency. For each fiscal year between 1966 and 2000, we rely on state-level military prime contract data constructed by [Nakamura and Steinsson \(2014\)](#). For the remaining sample period from 2001 until 2018, we use electronic DD Form 350 data available from www.USAspending.gov.

In order to identify the state-dependent spending multipliers, we add state-level changes in military spending interacted with indicator variables ($\mathbb{I}(\cdot)$), which provide information on US-states-years corresponding to the state of the economy. Note that the estimated multipliers with regional data are not directly comparable to the closed economy aggregate multipliers from the time series evidence in Section 5. The estimates from the regional analysis, open economy relative multipliers, measure the effect of an increase in government spending in one state relative to another state. However, these regional multipliers are useful in testing whether the effectiveness of fiscal policy depends on the US-state-differential conditions of the economy.

⁸Before 1995, [Nakamura and Steinsson \(2014\)](#) use state-level price indices constructed by [Del Negro \(1998\)](#) from 1969. After 1995, both papers by [Nakamura and Steinsson \(2014\)](#) and [Zidar \(2019\)](#) use county and metro level Cost of Living Index (COLI) published by the American Chamber of Commerce Researchers Association (ACCRA), later renamed as Council for Community and Economic Research (C2ER). As regional level COLI is designed to capture differences in price levels across regions within a year, [Nakamura and Steinsson \(2014\)](#) computed the state-level price indices by multiplying population-weighted COLI from the ACCRA for each state with the US aggregate CPI. We applied the same procedure to calculate the state-level price indices using the state-level COLI provided by [Zidar \(2019\)](#) and RPP from Census. There are a few missing US state-level inflation observations from Hawaii, Maine, New Hampshire, New Jersey, Rhode Island, and Vermont. We drop those US state-year observations if inflation data is missing.

We divide the state of economy based on the level of employment, inflation, and DNWR. The indicator variable for low employment, $\mathbb{I}(L(e_{it}))$ is one when the HP-filtered cyclical component of state-level employment to population ratio (e_{it}) is lower than 25th percentile of its distribution across US-states-and-years and zero otherwise. In addition, $\mathbb{I}(H(\pi_{it}))$ indicates high inflation US-states-years, which takes the value of one if biannual state-level inflation (π_{it}) is greater than 75th percentile of its distribution and zero otherwise. Lastly, the dummy variable $\mathbb{I}(H(DNWR_{it}))$ indicates US-states-years when more workers have the binding DNWR constraints. $\mathbb{I}(H(DNWR_{it}))$ is one when the biannual changes in the state-level differences between the share of workers with zero wage and the share of workers with wage cut is higher than 75% percentile from its distribution across states and years from 1979 to 2018.⁹

⁹Note that the regression specification does not include the level of state-level inflation and the cyclical component of employment themselves but contains dummy variables indicating a high inflation period or a low employment period. This specification is useful to avoid potential measurement errors in state-level measures of inflation, employment, and DNWR. For example, the level of inflation from our data set differs slightly from the one from [Hazell, Herreño, Nakamura, and Steinsson \(2020\)](#) but the indicator of high inflation is almost the same across the two. Our main results are also robust to using the [Hazell, Herreño, Nakamura, and Steinsson \(2020\)](#) state-level inflation data set.